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Chapter 1

Introduction

PLACEHOLDER

Chapter 2

Viscoelastic Fluid Mechanics and the Four-Roll Mill

The hydrodynamics of dilute and semi-dilute polymer solutions fundamentally departs from classical Newtonian fluid behavior. Macromolecular coils dissolved in a solvent undergo continuous conformational changes under applied flow fields, stretching along extensional trajectories and storing elastic energy with a characteristic relaxation memory. Consequently, the local stress state in polymeric fluids depends not merely on the instantaneous rate of deformation, but integrates the entire kinematic history experienced by fluid elements. Accurately modeling such transport phenomena requires a continuum mechanics formulation where universal conservation laws are coupled with non-linear, frame-indifferent constitutive equations.

In this chapter, we develop the continuum framework governing viscoelastic flows within the four-roll mill geometry. Starting from the foundational balance laws of mass and linear momentum, we introduce the kinematic reductions characteristic of confined, low-inertia polymeric flows. We then establish the constitutive formulation of viscoelasticity through the Upper-Convected Time Derivative, analyzing both the classical Oldroyd-B model, including the physical roots of its

extensional stress singularity, and non-linear stress-saturating closures such as the Giesekus and Phan-Thien-Tanner (PTT) formulations. Finally, we cast the governing system into dimensionless form using a viscous stress scale, detail the geometry and hyperbolic stagnation kinematics of the four-roll mill, and prescribe the boundary constraints that establish the boundary value problem.

2.1 General Conservation Laws

From a macroscopic continuum perspective, fluid motion is governed by the universal conservation laws of mass and linear momentum [1].

2.1.1 Mass Balance

For a fluid possessing local density $\rho(\boldsymbol{x}, t)$, the conservation of mass over an arbitrary control volume is expressed by the continuity equation:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \boldsymbol{u}) = 0. \tag{2.1}$$

2.1.2 Momentum Balance

Applying Newton's second law of motion to an infinitesimal fluid element yields the local momentum balance:

$$\rho \frac{\mathrm{D}\boldsymbol{u}}{\mathrm{D}t} = \nabla \cdot \boldsymbol{\sigma} + \rho \boldsymbol{g}, \tag{2.2}$$

where $\boldsymbol{g}$ represents external body force accelerations and $\boldsymbol{\sigma}$ denotes the symmetric total stress tensor. The left-hand side operator $\frac{\mathrm{D}}{\mathrm{D}t}$ is the substantial (or material) derivative, defined as:

$$\frac{\mathrm{D}\boldsymbol{u}}{\mathrm{D}t} \equiv \frac{\partial \boldsymbol{u}}{\partial t} + (\boldsymbol{u} \cdot \nabla)\boldsymbol{u}. \tag{2.3}$$

In the Eulerian framework, the substantial derivative expresses the total rate of velocity change experienced by a moving fluid parcel, including the time derivative $\frac{\partial \boldsymbol{u}}{\partial t}$ at a fixed spatial position and the convective part $(\boldsymbol{u} \cdot \nabla)\boldsymbol{u}$. Expanding Equation (2.2):

$$\rho \left(\frac{\partial \boldsymbol{u}}{\partial t} + \boldsymbol{u} \cdot \nabla \boldsymbol{u} \right) = \nabla \cdot \boldsymbol{\sigma} + \rho \boldsymbol{g}. \tag{2.4}$$

In order to separate isotropic hydrostatic effects from deviatoric shearing and stretching deformations, the total stress tensor $\boldsymbol{\sigma}$ is split into an isotropic pressure contribution and an extra-stress tensor $\boldsymbol{T}$:

$$\boldsymbol{\sigma} = -p\boldsymbol{I} + \boldsymbol{T}, \tag{2.5}$$

where p is the isotropic hydrodynamic pressure, $\boldsymbol{I}$ is the identity tensor, and $\boldsymbol{T}$ represents all viscous and viscoelastic stresses. Substituting Equation (2.5) into Equation (2.4):

$$\rho \left(\frac{\partial \boldsymbol{u}}{\partial t} + \boldsymbol{u} \cdot \nabla \boldsymbol{u} \right) = -\nabla p + \nabla \cdot \boldsymbol{T} + \rho \boldsymbol{g}. \tag{2.6}$$

2.2 Physical Assumptions

The general conservation equations can be simplified by introducing the scale-appropriate physical hypotheses of the four-roll mill:

1. **Incompressibility**: Under moderate operating pressures and low characteristic velocities, density variations are negligible ($\rho = \text{const}$). The continuity equation (2.1) simplifies to the divergence-free condition:

$$\nabla \cdot \boldsymbol{u} = \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0. \tag{2.7}$$

2. **Steady-state regime**: We focus on the fully established operating regime,

setting all explicit time derivatives to zero:

$$\frac{\partial(\cdot)}{\partial t} = \mathbf{0}. \tag{2.8}$$

3. **Absence of body forces**: The four-roll mill operates in a horizontal cross-sectional plane (x, y). Because gravity acts along the vertical cylinder axis (z), perpendicular to the flow plane, external body forces vanish: $\boldsymbol{g} = \mathbf{0}$.

4. **Isothermal operating conditions**: Because velocities are low and viscous heating is negligible, the temperature field remains uniform. Fluid transport properties, including viscosities and relaxation times, are therefore treated as constants.

5. **Two-dimensional planar kinematics**: Given the large aspect ratio of the cylindrical rollers along the z-axis, three-dimensional end effects are negligible in the transverse mid-plane. The velocity field is modeled as strictly planar:

$$\boldsymbol{u} = \begin{pmatrix} u(x,y) \\ v(x,y) \\ 0 \end{pmatrix}. \tag{2.9}$$

2.2.1 Stream Function Formulation

In two-dimensional incompressible flow, the solenoidal constraint $\nabla \cdot \boldsymbol{u} = 0$ can be satisfied directly by defining a scalar stream function $\psi(x, y)$ such that:

$$u = \frac{\partial \psi}{\partial y}, \qquad v = -\frac{\partial \psi}{\partial x}. \tag{2.10}$$

Because mixed partial derivatives commute ($\frac{\partial^2\psi}{\partial x\partial y} = \frac{\partial^2\psi}{\partial y\partial x}$), the velocity divergence vanishes identically:

$$\nabla \cdot \boldsymbol{u} = \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = \frac{\partial^2\psi}{\partial x\partial y} - \frac{\partial^2\psi}{\partial y\partial x} \equiv 0. \tag{2.11}$$

Parameterizing the velocity through ψ satisfies mass conservation everywhere by construction, replacing the two velocity components (u, v) with a single scalar field.

Under these physical assumptions and kinematically reduced velocity representation, the steady balance of linear momentum simplifies to:

$$\rho(\boldsymbol{u} \cdot \nabla\boldsymbol{u}) = -\nabla p + \nabla \cdot \boldsymbol{T}. \tag{2.12}$$

2.3 Constitutive Modeling of Viscoelastic Fluids

In two dimensions, the momentum balance (2.12) contains six unknown scalar fields $(u, v, p, T_{xx}, T_{xy}, T_{yy})$, while mass and momentum conservation provide only three equations. A constitutive equation for the extra-stress tensor $\boldsymbol{T}$ is required to close the system.

For dilute and semi-dilute polymer solutions, we split the extra-stress tensor $\boldsymbol{T}$ into a Newtonian solvent contribution and an elastic polymeric contribution $\boldsymbol{\tau}$ [1, 2]:

$$\boldsymbol{T} = 2\mu_s\boldsymbol{D} + \boldsymbol{\tau}, \tag{2.13}$$

where μ_s is the dynamic viscosity of the Newtonian solvent, and $\boldsymbol{D}$ is the symmetric rate-of-strain tensor:

$$\boldsymbol{D} = \frac{1}{2}\left[\nabla\boldsymbol{u} + (\nabla\boldsymbol{u})^T\right]. \tag{2.14}$$

Taking the divergence of the solvent stress and leveraging incompressibility ($\nabla \cdot \boldsymbol{u} = 0$):

$$\nabla \cdot (2\mu_s \boldsymbol{D}) = \mu_s \nabla \cdot \left(\nabla \boldsymbol{u} + (\nabla \boldsymbol{u})^T\right) = \mu_s \left(\nabla^2 \boldsymbol{u} + \nabla(\nabla \cdot \boldsymbol{u})\right) = \mu_s \nabla^2 \boldsymbol{u}. \quad (2.15)$$

Substituting Equation (2.15) into Equation (2.12) yields the steady momentum equation:

$$\rho(\boldsymbol{u} \cdot \nabla \boldsymbol{u}) = -\nabla p + \mu_s \nabla^2 \boldsymbol{u} + \nabla \cdot \boldsymbol{\tau}. \quad (2.16)$$

2.3.1 Upper-Convected Time Derivative

Polymer chains suspended in a solvent stretch, orient, and relax over finite timescales. Because fluid parcels translate, rotate, and deform as they move through the domain, constitutive relations require a rate of stress that is invariant under changes of reference frame (material frame-indifference). A standard time derivative evaluated at a fixed point would interpret rigid-body rotation or parcel stretching as a change in stress, even if the internal state of the polymer remained unaltered.

To measure true rheological evolution, the time rate of change must be evaluated in a local coordinate system that deforms affinely with the fluid element. This objective tensor rate is the **Upper-Convected Time Derivative** (UCTD), denoted by $\overset{\nabla}{\boldsymbol{\tau}}$:

$$\overset{\nabla}{\boldsymbol{\tau}} \equiv \frac{\partial \boldsymbol{\tau}}{\partial t} + \boldsymbol{u} \cdot \nabla \boldsymbol{\tau} - (\nabla \boldsymbol{u})\boldsymbol{\tau} - \boldsymbol{\tau}(\nabla \boldsymbol{u})^T, \quad (2.17)$$

where $(\nabla \boldsymbol{u})_{ij} = \partial u_i / \partial x_j$. For steady flows ($\partial \boldsymbol{\tau}/\partial t = \boldsymbol{0}$), the operator combines advective stress transport along streamlines, $\boldsymbol{u} \cdot \nabla \boldsymbol{\tau}$, with kinematic stretching and rotation induced by local velocity gradients, $-(\nabla \boldsymbol{u})\boldsymbol{\tau} - \boldsymbol{\tau}(\nabla \boldsymbol{u})^T$.

2.3.2 Oldroyd-B Constitutive Model

The Oldroyd-B model [3] provides the standard starting point for viscoelastic fluid modeling. It idealizes macromolecules as non-interacting Hookean dumbbells suspended in a Newtonian solvent [1]. Under steady conditions, the polymeric extra-stress tensor $\boldsymbol{\tau}$ satisfies:

$$\boldsymbol{\tau} + \lambda \overset{\nabla}{\boldsymbol{\tau}} = 2\mu_p \boldsymbol{D}, \tag{2.18}$$

where λ is the relaxation time and μ_p is the polymeric viscosity. The total zero-shear dynamic viscosity of the solution is:

$$\mu_0 = \mu_s + \mu_p. \tag{2.19}$$

In steady simple shear flow, Oldroyd-B predicts a constant viscosity $\eta(\dot{\gamma}) = \mu_0$, reproducing the behavior of Boger fluids.

Planar Extensional Behavior

In strong extensional flows, the Oldroyd-B model exhibits an unphysical stress singularity. For a steady planar elongational flow centered at the origin, $u = Ex$ and $v = -Ey$ with extension rate $E > 0$, the tensile normal stress along the stretching axis is:

$$\tau_{xx} = \frac{2\mu_p E}{1 - 2\lambda E}. \tag{2.20}$$

As the extension rate approaches the critical threshold:

$$E_{\text{crit}} = \frac{1}{2\lambda}, \tag{2.21}$$

the denominator vanishes. Because Hookean springs have infinite extensibility, the polymer chains can stretch without bound, driving both τ_{xx} and the exten-

sional viscosity η_E to infinity.

This stress singularity near $2\lambda E \to 1$ is the constitutive origin of the **High-Weissenberg Number Problem** (HWNP). Near extensional stagnation points, it generates steep stress layers that cause numerical instability in standard mesh-based solvers [4].

2.3.3 Giesekus Model

To overcome the unphysical stress singularity of Oldroyd-B, Giesekus [1, 5] introduced a quadratic non-linear stress contribution derived from configuration-dependent hydrodynamic drag. In semi-dilute solutions, stretching macromolecular chains align parallel to each other, reducing drag along the orientation direction relative to the transverse directions. This anisotropic friction is incorporated by modifying the stress relaxation rate:

$$\boldsymbol{\tau} + \lambda \overset{\nabla}{\boldsymbol{\tau}} + \frac{\alpha\lambda}{\mu_p}\boldsymbol{\tau}^2 = 2\mu_p \boldsymbol{D}, \tag{2.22}$$

where $\alpha \in [0, 0.5]$ is the dimensionless mobility parameter, and $\boldsymbol{\tau}^2 = \boldsymbol{\tau} \cdot \boldsymbol{\tau}$. When $\alpha = 0$, isotropic drag is recovered and the model reduces identically to Oldroyd-B. For $\alpha > 0$, the quadratic dissipation term $\boldsymbol{\tau}^2$ provides two key rheological enhancements:

1. **Bounded extensional plateau**: Extensional stresses no longer diverge at E_{crit}; instead, η_E reaches an asymptotic finite plateau at high strain rates, completely resolving the constitutive singularity.

2. **Shear-thinning behavior**: The steady shear viscosity $\eta(\dot{\gamma})$ decreases monotonically with increasing shear rate, closely matching experimental polymer melt and concentrated solution data.

2.3.4 Linear Phan-Thien-Tanner (PTT) Model

Derived from the transient network theory of concentrated polymer solutions and melts [1, 6], the Phan-Thien-Tanner (PTT) model treats macromolecules as a network of temporary entanglements. Under strong deformation, mechanical tension on the network strands accelerates junction destruction. In its linear formulation, this stress-induced network breakdown is modeled by scaling the linear relaxation term:

$$f(\boldsymbol{\tau})\boldsymbol{\tau} + \lambda \overset{\nabla}{\boldsymbol{\tau}} = 2\mu_p \boldsymbol{D}, \tag{2.23}$$

where the scalar function $f(\boldsymbol{\tau})$ is defined as:

$$f(\boldsymbol{\tau}) = 1 + \frac{\varepsilon\lambda}{\mu_p}\mathrm{tr}(\boldsymbol{\tau}), \tag{2.24}$$

with $\mathrm{tr}(\boldsymbol{\tau}) = \tau_{xx} + \tau_{yy}$ and $\varepsilon \geq 0$ representing the dimensionless extensibility parameter. When $\varepsilon = 0$, $f(\boldsymbol{\tau}) \equiv 1$ and the linear Oldroyd-B constitutive law is recovered. For $\varepsilon > 0$, high trace stresses accelerate stress relaxation, bounding extensional viscosity growth and preventing unbounded coil stretch near extensional stagnation points.

2.4 Nondimensionalization and Governing Equa< tions

Nondimensionalization scales physical variables to order unity and establishes the governing dimensionless groups [4]. In low-Reynolds viscoelastic flows, scaling by a viscous stress unit rather than an inertial dynamic pressure improves numerical conditioning and balances the magnitudes of the governing differential operators.

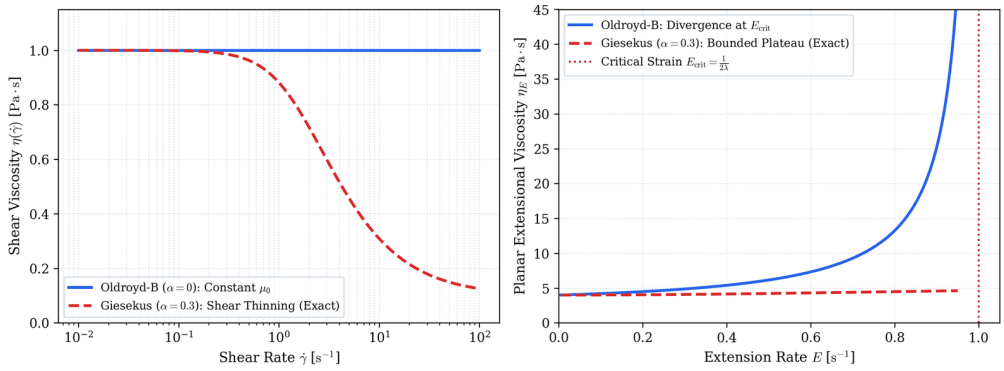


Figure 2.1: Example comparison of rheological model responses: (a) steady shear viscosity $\eta(\dot{\gamma})$ as a function of shear rate, illustrating the constant Newtonian-like plateau of Oldroyd-B ($\alpha = 0$) versus the non-linear shear-thinning response of Giesekus ($\alpha > 0$); (b) planar extensional viscosity $\eta_E(E)$ as a function of extension rate, contrasting the unphysical asymptotic divergence of Oldroyd-B at the critical strain threshold $E_{\text{crit}} = 1/(2\lambda)$ with the bounded saturation plateau provided by non-linear closures.

2.4.1 Reference Scales and Dimensionless Parameters

We choose reference length H_{ref} (the roller radius R), reference velocity U_{ref} (the roller surface peripheral speed u_{roll}), and a characteristic dynamic viscosity scale η_0 (taken as the baseline total viscosity μ_0). The dimensionless variables are defined as:

$$\boldsymbol{x}^* = \frac{\boldsymbol{x}}{H_{\text{ref}}}, \quad \boldsymbol{u}^* = \frac{\boldsymbol{u}}{U_{\text{ref}}}, \quad p^* = \frac{p}{\frac{\eta_0 U_{\text{ref}}}{H_{\text{ref}}}}, \quad \boldsymbol{\tau}^* = \frac{\boldsymbol{\tau}}{\frac{\eta_0 U_{\text{ref}}}{H_{\text{ref}}}}. \tag{2.25}$$

Dropping asterisks for notational simplicity, the flow regime is governed by three fundamental dimensionless groups:

$$Re = \frac{\rho U_{\text{ref}} H_{\text{ref}}}{\eta_0} \qquad \text{(Reynolds number)}, \tag{2.26}$$

$$Wi = \frac{\lambda U_{\text{ref}}}{H_{\text{ref}}} \qquad \text{(Weissenberg number)}, \tag{2.27}$$

$$\beta = \frac{\mu_s}{\mu_0} = \frac{\mu_s}{\mu_s + \mu_p} \qquad \text{(Viscosity ratio)}. \tag{2.28}$$

Each dimensionless group carries a precise physical meaning:

Reynolds number (Re): Compares convective inertial forces to viscous dissipation forces. In the present benchmark, $Re \ll 1$, placing the system firmly in the laminar creeping flow (Stokes) regime.

Weissenberg number (Wi): Compares the characteristic elastic relaxation time of the polymer λ to the characteristic kinematic timescale of the flow $H_{\text{ref}}/U_{\text{ref}}$. It quantifies the degree of flow-induced macromolecular orientation and elastic non-linearity.

Viscosity ratio (β): Measures the fraction of total viscosity contributed by the Newtonian solvent. The limits $\beta = 1$ and $\beta \to 0$ correspond to a purely Newtonian fluid and a solvent-free polymer melt, respectively. The dimensionless solvent and polymeric viscosities are given by $\mu_s^* = \beta$ and $\mu_p^* = 1 - \beta$.

2.4.2 Dimensionless Governing System

Expressed in terms of dimensionless variables, the steady planar governing equations become:

$$\nabla \cdot \boldsymbol{u} = 0, \tag{2.29}$$

$$Re(\boldsymbol{u} \cdot \nabla \boldsymbol{u}) + \nabla p - \beta \nabla^2 \boldsymbol{u} - \nabla \cdot \boldsymbol{\tau} = \boldsymbol{0}, \tag{2.30}$$

$$\boldsymbol{\mathcal{G}}(\boldsymbol{\tau}) + Wi\left(\boldsymbol{u} \cdot \nabla \boldsymbol{\tau} - (\nabla \boldsymbol{u})\boldsymbol{\tau} - \boldsymbol{\tau}(\nabla \boldsymbol{u})^T\right) - 2(1-\beta)\boldsymbol{D} = \boldsymbol{0}, \tag{2.31}$$

where the constitutive operator $\boldsymbol{\mathcal{G}}(\boldsymbol{\tau})$ provides a unified closure for the models under investigation:

$$\boldsymbol{\mathcal{G}}(\boldsymbol{\tau}) = \begin{cases} \boldsymbol{\tau} & \text{(Oldroyd-B)}, \\ \boldsymbol{\tau} + \frac{\alpha Wi}{1-\beta}\boldsymbol{\tau}^2 & \text{(Giesekus)}, \\ \left(1 + \frac{\varepsilon Wi}{1-\beta}\mathrm{tr}(\boldsymbol{\tau})\right)\boldsymbol{\tau} & \text{(Linear PTT)}. \end{cases} \tag{2.32}$$

2.5 Four-Roll Mill

The four-roll mill is a classic benchmark designed to generate controlled two-dimensional planar extensional flow with a central stagnation point [7–10]. Originally introduced by Taylor [7] and systematically analyzed by Fuller and Leal [8,9], it provides an ideal apparatus for investigating macromolecular deformation away from solid walls. As illustrated conceptually in Figure 2.2, the setup comprises four cylindrical rollers placed symmetrically inside a closed square cavity.

2.5.1 Flow Index Parameter

In the central region bounded by the four rollers, the velocity field can be represented as a linear superposition of extension and solid-body rotation [8, 9]:

$$\boldsymbol{u}(x, y) = \begin{pmatrix} Ex + \omega y \\ -Ey - \omega x \end{pmatrix}, \tag{2.33}$$

where $E \geq 0$ is the extension rate and $\omega \geq 0$ is the angular rotation rate. To quantify the relative magnitude of extension versus rotation, we define a dimensionless flow type parameter $\chi \in [-1, 1]$:

$$\chi = \frac{E - \omega}{E + \omega}. \tag{2.34}$$

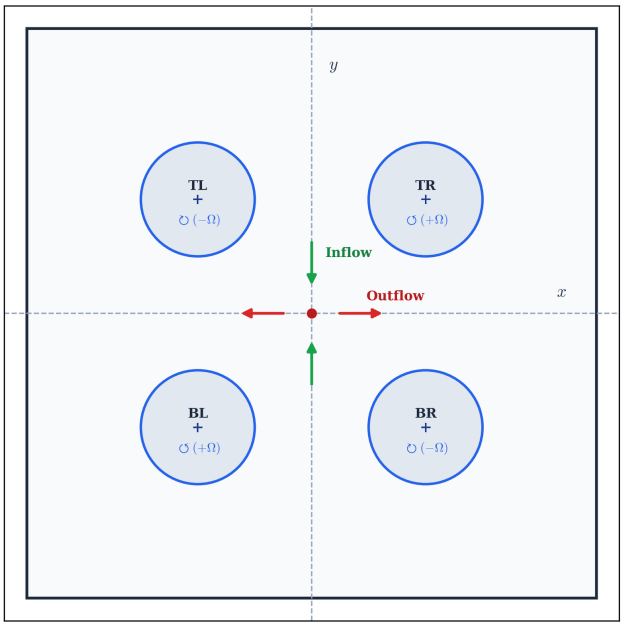


Figure 2.2: Conceptual diagram of the four-roll mill geometry, illustrating the four rotating cylinders (adjacent rolls counter-rotate with angular velocity Ω, diagonal rolls co-rotate), the central hyperbolic stagnation point $(0, 0)$, and the resulting elongational inflow and outflow directions.

Depending on roller rotation speeds and directions, this local kinematics covers three canonical flow regimes (in Figure 2.3):

$\chi = +1$ $(\omega = 0)$: **Pure Planar Extensional Flow**. Generated when adjacent rollers counter-rotate symmetrically (diagonal rollers co-rotate). Fluid elements experience pure elongation along x and compression along y, producing hyperbolic streamlines $xy = \text{const}$.

$\chi = 0$ $(E = \omega)$: **Simple Shear Flow**. Characterized by equal strain and rotation rates, producing rectilinear shear streamlines locally near the origin.

$\chi = -1$ $(E = 0)$: **Pure Solid-Body Rotation**. Generated when rollers rotate in unison to drive bulk cavity circulation, resulting in concentric circular streamlines without stretching.

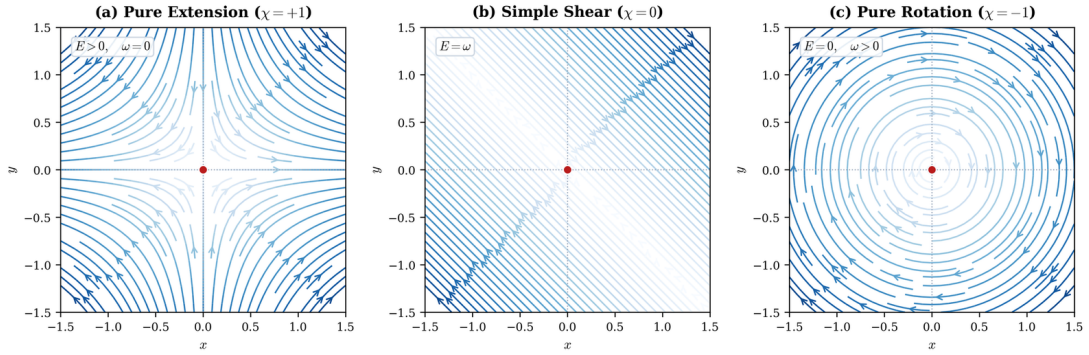


Figure 2.3: Streamline patterns of the two-dimensional linear flow field in Equation (2.33) for the three canonical regimes of the flow classification parameter χ: (a) pure planar extension ($\chi = +1$), (b) simple shear ($\chi = 0$), and (c) pure solid-body rotation ($\chi = -1$). The red marker indicates the central stagnation point $(0, 0)$.

2.5.2 Stagnation Point

In the planar extensional regime ($\chi = +1$), the velocity vanishes identically at the cavity center:

$$\boldsymbol{u}(0, 0) = \mathbf{0}. \tag{2.35}$$

Along the vertical inflow streamline ($x = 0$), incoming fluid parcels decelerate toward zero velocity, reaching asymptotically long residence times near the origin. This kinematics allows elastic strain to accumulate over durations comparable to or exceeding the polymer relaxation time λ, making the four-roll mill an ideal benchmark for probing strong extensional stresses far from solid walls.

2.5.3 Geometric Setup and Operating Parameters

Following the established benchmark setup of D'Avino et al. [11], we define the physical dimensions and operating conditions of the apparatus:

Square cavity: Side length $L = 50\,\text{mm}$ spanning the domain $[-25, 25]\,\text{mm}$.

Cylindrical rollers: Radius $R = 5\,\text{mm}$ ($H_{\text{ref}} = R = 0.1L$).

Roller centers: Located symmetrically at $(\pm 10, \pm 10)$ mm ($d_{\text{roll}} = 10\,\text{mm}$).

Roller surface speed: Peripheral velocity $u_{\text{roll}} = \Omega R = 8.333 \times 10^{-3}\,\text{m/s}$.

Dimensionless domain: In units of $H_{\text{ref}} = R$, the computational domain spans $[-5, 5]^2$, with roller centers located at $(\pm 2.0, \pm 2.0)$ and non-dimensional roller radius $R^* = 1.0$.

As shown in Figure 2.4, these dimensions define the operational space and solid boundaries of the cavity.

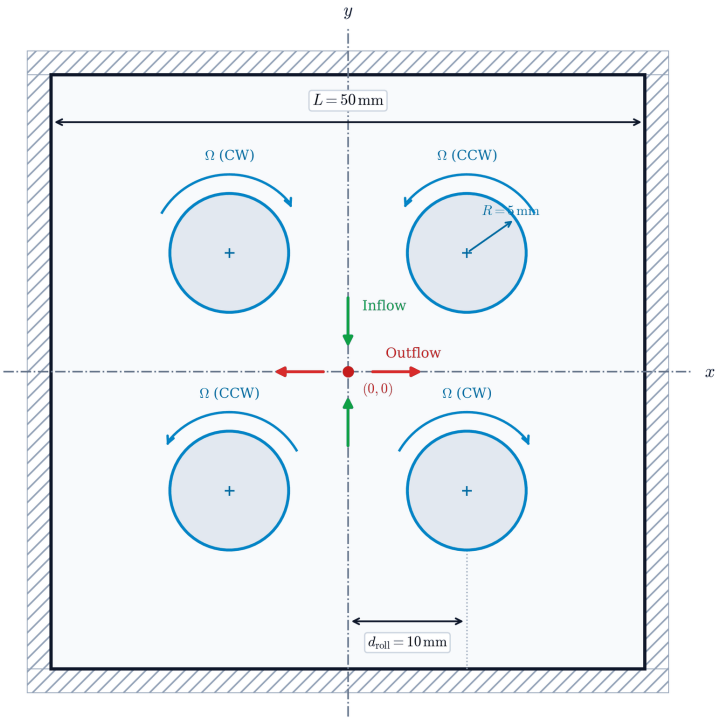


Figure 2.4: Detailed geometric dimensions and coordinate setup of the four-roll mill: square cavity side $L = 50\,\text{mm}$, cylinder radii $R = 5\,\text{mm}$, roller center offset $d_{\text{roll}} = 10\,\text{mm}$, Cartesian axes (x, y), and peripheral rotation directions with angular velocity Ω.

The geometric, fluid physical, and characteristic dimensionless parameters investigated in this work are organized systematically in Table 2.1.

Table 2.1: Geometric, physical, and dimensionless parameters of the four-roll mill benchmark.

Parameter	Symbol	Value / Range	Unit
Geometric and Kinematic Setup			
Cavity side length	L	0.05	m
Roller radius (reference length)	$H_{\text{ref}} = R$	0.005	m
Roller center offset	d_{roll}	0.010	m
Roller peripheral velocity	$u_{\text{roll}} = U_{\text{ref}}$	8.333×10^{-3}	m/s
Fluid Physical Properties			
Fluid density	ρ	1000	kg/m^3
Characteristic viscosity scale	$\eta_0 = \mu_0$	1.0	$\text{Pa} \cdot \text{s}$
Solvent dynamic viscosity	μ_s	0.1–0.5	$\text{Pa} \cdot \text{s}$
Polymer dynamic viscosity	μ_p	0.5–0.9	$\text{Pa} \cdot \text{s}$
Polymer relaxation time	λ	0.05–1.0	s
Dimensionless Governing Groups			
Reynolds number	Re	$0.0417 \ll 1$	–
Weissenberg number	Wi	0.083–1.667	–
Viscosity ratio	β	0.1–0.5	–

2.5.4 Boundary Conditions

To complete the mathematical formulation, boundary conditions are imposed along allrelevant domain boundaries:

1. **No-Slip on Roller Surfaces (Γ_{rolls})**: On the boundary of each cylinder $k \in \{\text{TL}, \text{TR}, \text{BL}, \text{BR}\}$, the fluid velocity satisfies the no-slip Dirichlet condition with specified angular rotation:

$$\boldsymbol{u}|_{\Gamma_k} = \boldsymbol{\Omega}_k \times (\boldsymbol{x} - \boldsymbol{x}_k), \tag{2.36}$$

   where $\boldsymbol{x}_k$ is the roller center. In the planar extensional regime ($\chi = +1$), top-left (TL) and bottom-right (BR) cylinders rotate clockwise ($-\Omega$), while top-right (TR) and bottom-left (BL) cylinders rotate counter-clockwise ($+\Omega$).

2. **No-Slip on Walls (Γ_{outer})**: Along the stationary external boundaries of

the square cavity, the velocity vanishes identically:

$$\boldsymbol{u}|_{\Gamma_{\text{outer}}} = \mathbf{0}. \tag{2.37}$$

3. **Pressure Gauge Fixing**: Because pressure appears in the momentum balance (2.30) exclusively through its spatial gradient ∇p, the pressure field is determined only up to an arbitrary additive constant ($p \to p + C$) under pure velocity Dirichlet conditions. To eliminate this degree of freedom, pressure is anchored with a single reference point:

$$p(\boldsymbol{x}_0) = 0. \tag{2.38}$$

Chapter 3

Physics-Informed Neural Networks in Complex Rheology

Feedforward deep neural networks, commonly known as Multi-Layer Perceptrons (MLPs), provide flexible continuous approximations for non-linear functions [12]. An MLP is formed by composing a sequence of parametric functions, called layers. Each layer maps its inputs through a linear transformation followed by a non-linear operation. With these layers in series, the network can represent continuous functional mappings of arbitrary complexity without requiring explicit basis functions.

In scientific computing, neural networks represent physical fields directly as continuous functions over spatial coordinates, rather than storing discrete state variables on a mesh. However, using purely data-driven networks for fluid mechanics presents clear limits: they require large amounts of labeled data, can violate conservation laws, and cannot infer latent material properties on their own.

Physics-Informed Neural Networks (PINNs) address these limits by embedding governing partial differential equations directly into the training loss. The differential system formulated in Chapter 2, governed by mass conservation, vis-

coelastic momentum balance, and frame-indifferent constitutive transport, represents a coupled set of non-linear equations that poses well-known challenges to traditional mesh-based methods. In this chapter, we formulate this continuous boundary-value problem within a Scientific Machine Learning (SciML) framework. By embedding momentum and constitutive balances into the optimization objective through automatic differentiation, the PINN framework enables both forward state simulation and inverse rheological parameter discovery without spatial discretization meshes.

3.1 Fundamentals of Deep Learning

Deep learning models approximate continuous mathematical functions by learning parameters directly from data [12]. In this section, we review the fundamental building blocks of feedforward networks: the artificial neuron, activation functions, multi-layer architectures, loss functions, optimization algorithms, and gradient computation.

3.1.1 The Artificial Neuron and Non-Linear Activation Functions

The elementary computational unit of a neural network is the artificial neuron. Given an input vector $\boldsymbol{x}$, a neuron computes a linear combination of the inputs followed by a non-linear activation:

$$z = \boldsymbol{w} \cdot \boldsymbol{x} + b, \qquad a = \sigma(z), \tag{3.1}$$

where $\boldsymbol{w}$ is the weight vector, b is a scalar bias, $\sigma(\cdot)$ is the activation function and a is the output. Non-linear activations are essential: without them, composing multiple linear layers simply collapses into a single linear transformation.

In classical machine learning, piecewise-linear functions such as the Rectified Linear Unit, $\mathrm{ReLU}(z) = \max(0, z)$, are widely favored for their computational efficiency. However, in scientific computing and fluid mechanics, the network outputs parameterize continuous physical fields governed by differential equations. These governing balance laws involve spatial derivatives of first, second, or third order, such as viscous diffusion $\mu_s \nabla^2 \boldsymbol{u}$ or stress divergence $\nabla \cdot \boldsymbol{\tau}$.

Because the first derivative of ReLU is discontinuous and its second derivative is zero almost everywhere, evaluating a ReLU network inside differential residuals eliminates diffusive and curvature terms. Physics-informed models therefore require smooth, differentiable activation functions ($\mathcal{C}^2$ or higher). In this work, we employ the Sigmoid Linear Unit (SiLU, also known as Swish) [12]:

$$\mathrm{SiLU}(z) = \frac{z}{1 + e^{-z}}. \tag{3.2}$$

The SiLU activation is smooth and non-monotonic. Unlike the hyperbolic tangent (tanh), which saturates toward constant values for large positive inputs, SiLU remains active in the positive regime. This property maintains healthy gradient propagation in flow regions with intense shear and steep localized stress gradients, which are characteristic of viscoelastic flows.

3.1.2 Multi-Layer Perceptron Architecture

By arranging artificial neurons into sequential layers, we construct a Multi-Layer Perceptron (MLP). The network maps an input vector $\boldsymbol{x}$ to an output prediction

$\hat{\boldsymbol{u}}$ by propagating information through L successive layers [12]:

$$\begin{aligned} \boldsymbol{h}^{(1)} &= \sigma\left(\boldsymbol{W}^{(1)}\boldsymbol{x} + \boldsymbol{b}^{(1)}\right), \\ \boldsymbol{h}^{(l)} &= \sigma\left(\boldsymbol{W}^{(l)}\boldsymbol{h}^{(l-1)} + \boldsymbol{b}^{(l)}\right), \quad \text{for } l = 2, \ldots, L-1, \\ \hat{\boldsymbol{u}} &= \boldsymbol{W}^{(L)}\boldsymbol{h}^{(L-1)} + \boldsymbol{b}^{(L)}, \end{aligned} \tag{3.3}$$

where $\boldsymbol{h}^{(l)}$ denotes the hidden feature vector at layer l, and $\boldsymbol{W}^{(l)}$ and $\boldsymbol{b}^{(l)}$ represent the weight matrix and bias vector of that layer. In our implementation, the inputs $\boldsymbol{x} = (x, y)$ are spatial coordinates within the fluid cavity, while the outputs $\hat{\boldsymbol{u}}$ parameterize the physical state variables: the stream function ψ, hydrodynamic pressure p, and extra-stress tensor components $\boldsymbol{\tau}$.

A central theoretical property of MLPs is universal function approximation [13,14]: a feedforward network with sufficient width and smooth non-linear activations can approximate any continuous function on a compact domain to arbitrary precision. However, universal approximation guarantees only the existence of a suitable set of weights, not that standard gradient-based optimization will automatically find them or that spatial derivatives will match the physical solution with equal fidelity.

Importance of Weight Initialization

The initialization of network parameters strongly conditions the resulting optimization dynamics. Because deep architectures compose multiple transformations and non-linearities, inappropriate initial weight magnitudes can cause layer activations and backpropagated gradients to either vanish or explode across depth [12]. In scientific computing and differential problems, where non-linear convective terms and tensor couplings are evaluated, uncalibrated initial weights can trigger extreme initial residual spikes that immediately destabilize gradient

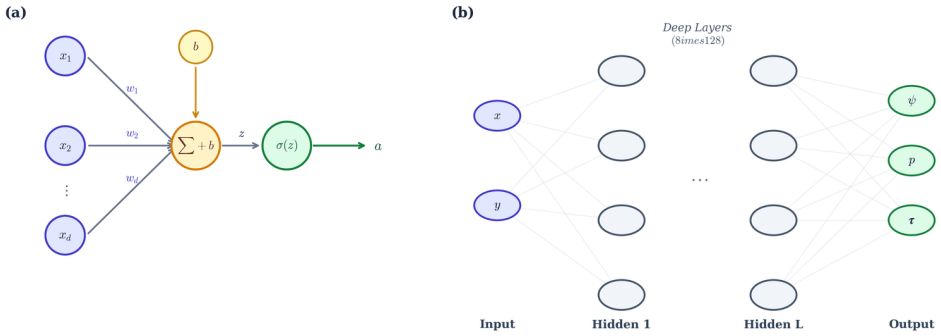


Figure 3.1: Neural network representations: (a) schematic of an artificial neuron computing an affine combination $z = \boldsymbol{w} \cdot \boldsymbol{x} + b$ followed by a smooth activation $\sigma(z)$; (b) deep Multi-Layer Perceptron (MLP) architecture propagating coordinate inputs (x, y) through L hidden layers to parameterize continuous physical fields $(\psi, p, \boldsymbol{\tau})$.

descent. Appropriate variance-preserving initialization schemes, such as scaling parameter distributions inversely with layer widths, help preserve stable gradient flow and avoid optimization divergence. Specific configuration choices regarding hidden and output layer initialization adopted in this work are detailed in Chapter 4.

3.1.3 Loss Functions and Optimization Algorithms

Training an MLP consists of identifying a parameter set $\boldsymbol{\theta} = \{\boldsymbol{W}^{(l)}, \boldsymbol{b}^{(l)}\}$ that minimizes a scalar objective function $\mathcal{L}(\boldsymbol{\theta})$. In supervised regression, the discrepancy between model predictions and target reference measurements is commonly quantified through the Mean Squared Error (MSE):

$$\mathcal{L}_{\text{MSE}}(\boldsymbol{\theta}) = \frac{1}{N} \sum_{i=1}^{N} \left\| \hat{\boldsymbol{u}}(\boldsymbol{x}_i; \boldsymbol{\theta}) - \boldsymbol{u}_i^{\text{target}} \right\|_2^2 . \tag{3.4}$$

Parameters are iteratively updated along descent directions derived from the loss gradient:

$$\boldsymbol{\theta} \leftarrow \boldsymbol{\theta} - \eta \boldsymbol{d}_k, \tag{3.5}$$

where $\eta > 0$ represents the learning rate (step size) and $\boldsymbol{d}_k$ denotes the search direction at iteration k.

In scientific machine learning, gradient-based solvers must navigate non-convex loss surfaces characterized by ill-conditioned curvature and local minima. Two widely adopted algorithms in this setting are:

1. First-order adaptive optimization (Adam): The Adaptive Moment Estimation algorithm [15] computes individual adaptive learning rates for each parameter by maintaining exponentially decaying running averages of past gradients (first moment) and squared gradients (uncentered second moment). This moment-based scaling renders Adam particularly robust against ill-conditioned curvature and saddle points, making it highly effective for early-stage exploration across rugged non-convex landscapes.

2. Second-order quasi-Newton optimization (L-BFGS): The Limited-memory Broyden–Fletcher–Goldfarb–Shanno algorithm [16] incorporates local curvature information by constructing low-rank approximations of the inverse Hessian matrix using the history of recent gradient evaluations. While higher-order computation requires evaluation over full data batches, L-BFGS provides deterministic superlinear convergence in the vicinity of local minima, capable of driving differential equation residuals to strict numerical tolerances.

Both algorithms share the common objective of driving the objective function toward an optimal stationary point, yet they exhibit distinct trade-offs: Adam offers robust stochastic exploration with modest per-iteration memory requirements, whereas L-BFGS delivers deterministic precision and rapid asymptotic convergence at the expense of curvature memory buffers.

3.1.4 Backpropagation

To update network parameters via gradient descent, we compute the partial derivatives of the scalar loss with respect to all weights and biases, $\nabla_{\boldsymbol{\theta}}\mathcal{L}$. Evaluating these gradients through individual numerical perturbations would be computationally prohibitive.

Instead, neural networks compute analytical gradients using the **Automatic Differentiation (AD)** algorithm [17]. Automatic differentiation evaluates derivatives to machine precision by applying the chain rule to elementary operations along the computational graph. In standard machine learning, reverse-mode AD is widely known as the **backpropagation** algorithm, primarily employed to evaluate the gradient of the scalar loss with respect to network parameters.

This reverse-mode automatic differentiation evaluates all parameter gradients simultaneously at a computational cost comparable to a single forward pass, making deep network training feasible.

3.2 Neural Networks: Capabilities and Physical Limitations

3.2.1 Key Advantages

Neural networks offer distinctive characteristics that make them attractive for continuous physical modeling:

1. **High-Dimensional Functional Approximation**: The universal approximation capacity allows neural networks to represent complex, coupled physical fields (velocities, pressures, and stress tensors) within a unified functional representation, without requiring specialized interpolation elements for each variable.

2. **Meshless Representation**: Traditional grid-based methods (Finite Element, Finite Volume, and Finite Difference Methods) represent state variables exclusively at discrete vertices or cell centers, relying on explicit mesh connectivity. In contrast, a neural network provides a continuous, globally defined mapping $\boldsymbol{x} \mapsto \hat{\boldsymbol{u}}(\boldsymbol{x})$ over the entire domain, capable of being evaluated at arbitrary spatial coordinates. However, being meshless does not imply freedom from geometric considerations: defining the domain, enforcing boundary conditions, and designing effective collocation point distributions remain essential and influential steps.

3. **Hardware Acceleration and Di□erentiability**: Modern deep learning frameworks are built upon dense linear algebra operations optimized for parallel accelerator hardware (GPUs and TPUs), and provide built-in automatic differentiation for computing spatial derivatives of the represented fields.

3.2.2 Limits of Purely Data-Driven Machine Learning

Despite their high expressivity, deploying purely data-driven black-box networks to fluid mechanics and non-Newtonian rheology reveals fundamental limitations:

Data scarcity in experimental rheology: Deep neural networks typically rely on large labeled datasets. In fluid experiments, acquiring dense velocity fields via Particle Image Velocimetry (PIV) is labor-intensive, while non-intrusive measurements of internal extra-stress components are very hard to obtain in complex geometries.

Absence of physical grounding: Purely data-driven models minimize empirical loss on discrete observations without physical constraints. Consequently, their predictions often violate incompressibility ($\nabla \cdot \boldsymbol{u} \neq 0$), mo-

mentum balance, and constitutive laws, generating non-physical artifacts when interpolating or extrapolating.

Overfitting: In sparse-data regimes, over-parameterized networks can easily fit measurement noise, obtaining low training errors while losing accuracy in unmonitored regions.

Inability to discover governing parameters: Black-box models learn input-output mappings, but cannot extract underlying physical properties, such as relaxation times or polymeric viscosities, that govern flow dynamics.

These limitations show that data alone are insufficient for scientific modeling in rheology, motivating frameworks that embed physical conservation laws directly into the loss function.

3.3 Physics-Informed Neural Network Framework

The Physics-Informed Neural Network (PINN) framework, formalized by Raissi, Perdikaris, and Karniadakis [18], addresses the problems of purely data-driven modeling by incorporating differential partial equations directly into the network training objective.

3.3.1 Physics-Constrained Loss Formulation

Following the unified boundary-value problem (BVP) notation established in the scientific machine learning literature [19], consider a physical state described by a vector of continuous fields $\boldsymbol{u}(\boldsymbol{x})$ defined over a spatial domain $\Omega \subset \mathbb{R}^d$ bounded

by $\partial\Omega$:

$$\mathcal{F}[\boldsymbol{u}(\boldsymbol{x});\boldsymbol{\lambda}] = \mathbf{0}, \quad \boldsymbol{x} \in \Omega, \tag{3.6}$$

$$\mathcal{B}[\boldsymbol{u}(\boldsymbol{x})] = \boldsymbol{g}(\boldsymbol{x}), \quad \boldsymbol{x} \in \partial\Omega, \tag{3.7}$$

where:

$\boldsymbol{u}(\boldsymbol{x})$ denotes the physical state variables (in our viscoelastic framework, $\boldsymbol{u} = (\psi, p, \boldsymbol{\tau})$, comprising stream function, pressure, and extra-stress tensor components);

$\mathcal{F}$ is a general non-linear differential operator encoding the governing field equations (mass conservation, momentum balance, and constitutive transport equations);

$\boldsymbol{\lambda}$ denotes physical or constitutive parameters governing the flow dynamics (such as solvent viscosity, polymeric viscosity, and relaxation time);

$\mathcal{B}$ denotes the boundary operator enforcing kinematic constraints on the boundaries $\partial\Omega$ (such as no-slip conditions along solid walls and prescribed angular velocities on roller surfaces), with prescribed boundary values $\boldsymbol{g}(\boldsymbol{x})$.

In a PINN, the continuous solution fields are parameterized by a deep neural network surrogate $\hat{\boldsymbol{u}}(\boldsymbol{x};\boldsymbol{\theta})$ parameterized by weights and biases $\boldsymbol{\theta}$. Evaluating the governing differential operator on the network outputs via automatic differentiation defines the pointwise differential residual:

$$\boldsymbol{f}(\boldsymbol{x};\boldsymbol{\theta},\boldsymbol{\lambda}) \equiv \mathcal{F}[\hat{\boldsymbol{u}}(\boldsymbol{x};\boldsymbol{\theta});\boldsymbol{\lambda}]. \tag{3.8}$$

The network parameters $\boldsymbol{\theta}$, alongside any unknown physical parameters $\boldsymbol{\lambda}$ in inverse problems, are optimized simultaneously by minimizing a composite multi-

objective loss:

$$\mathcal{L}_{\text{total}}(\boldsymbol{\theta}, \boldsymbol{\lambda}) = w_{\text{data}}\mathcal{L}_{\text{data}} + w_{\text{bc}}\mathcal{L}_{\text{bc}} + w_{\text{pde}}\mathcal{L}_{\text{pde}}, \tag{3.9}$$

where each individual loss component is evaluated as:

$$\mathcal{L}_{\text{data}} = \frac{1}{N_{\text{data}}} \sum_{i=1}^{N_{\text{data}}} \left\| \hat{\boldsymbol{u}}(\boldsymbol{x}_i^{\text{data}}) - \boldsymbol{u}_i^{\text{data}} \right\|_2^2, \tag{3.10}$$

$$\mathcal{L}_{\text{bc}} = \frac{1}{N_{\text{bc}}} \sum_{j=1}^{N_{\text{bc}}} \left\| \mathcal{B}[\hat{\boldsymbol{u}}(\boldsymbol{x}_j^{\text{bc}})] - \boldsymbol{g}(\boldsymbol{x}_j^{\text{bc}}) \right\|_2^2, \tag{3.11}$$

$$\mathcal{L}_{\text{pde}} = \frac{1}{N_{\text{pde}}} \sum_{k=1}^{N_{\text{pde}}} \left\| \boldsymbol{f}(\boldsymbol{x}_k^{\text{pde}}; \boldsymbol{\theta}, \boldsymbol{\lambda}) \right\|_2^2. \tag{3.12}$$

Here, $\{\boldsymbol{x}_i^{\text{data}}\}$ denotes available empirical observations or measurement locations, $\{\boldsymbol{x}_j^{\text{bc}}\}$ represents points sampled along physical boundaries $\partial\Omega$, and $\{\boldsymbol{x}_k^{\text{pde}}\}$ designates interior **collocation points** scattered across Ω.

This formulation naturally supports two complementary problem settings:

Forward problems: The physical parameters $\boldsymbol{\lambda}$ are known. The network optimizes its weights $\boldsymbol{\theta}$ to solve for the continuous fields $\hat{\boldsymbol{u}}(\boldsymbol{x})$, acting as a meshless continuous PDE solver.

Inverse problems: One or more constitutive parameters in $\boldsymbol{\lambda}$ are unknown. The network simultaneously reconstructs the unobserved latent fields and identifies the unknown parameters by combining empirical velocity observations $\mathcal{L}_{\text{data}}$ with physical consistency constraints $\mathcal{L}_{\text{pde}}$.

A key methodological principle of this thesis is the distinction between field reconstruction accuracy and parameter identifiability. A network may fit observed velocity fields accurately, yet inferred rheological parameters may remain uncertain if the kinematics lack sufficient variation or if the loss landscape exhibits degeneracies. Embedding physical equations provides essential structure,

but does not automatically guarantee unique parameter recovery.

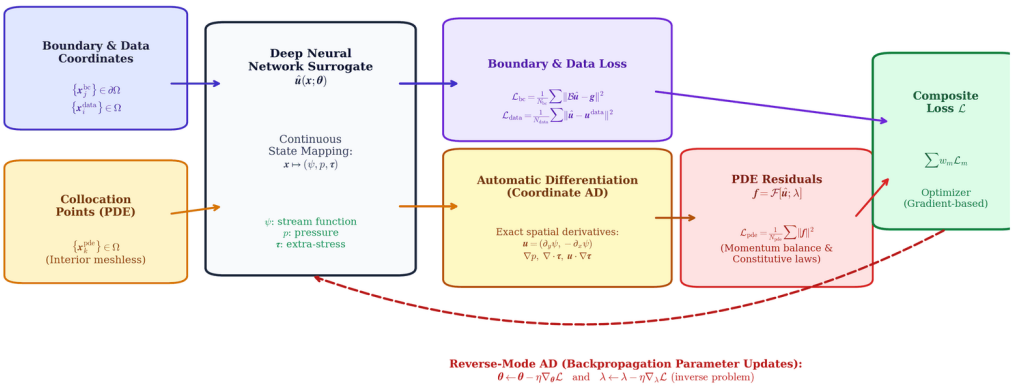


Figure 3.2: Schematic computational workflow of a Physics-Informed Neural Network (PINN): boundary coordinates $\{\boldsymbol{x}_j^{\text{bc}}\}$ and measurements $\{\boldsymbol{x}_i^{\text{data}}\}$ evaluate boundary adherence ($\mathcal{L}_{\text{bc}}$) and empirical matching ($\mathcal{L}_{\text{data}}$) (top branch), while interior collocation points $\{\boldsymbol{x}_k^{\text{pde}}\}$ pass through Coordinate Automatic Differentiation (AD) to compute exact spatial derivatives and differential PDE residuals ($\mathcal{L}_{\text{pde}}$, bottom branch). The composite loss $\mathcal{L}_{\text{total}}$ drives gradient descent via reverse-mode backpropagation, jointly optimizing surrogate weights $\boldsymbol{\theta}$ and identifying unknown physical parameters $\boldsymbol{\lambda}$ in inverse settings.

3.3.2 PINN specific features

Several operational features distinguish physics-informed training from conventional supervised learning:

1. **Training as a boundary-value solver**: In classical machine learning, a model is trained on a sample set to generalize to unseen statistical distributions. In PINNs, **training is the numerical solver**. The network is not designed to predict flows across arbitrary unseen geometries; rather, optimization directly solves the specific problem on the specified domain Ω. Available measurements $\mathcal{L}_{\text{data}}$ act as an active data anchor that guides the network toward the true physical state.

2. **Physical regularization and dynamic loss weighting**: In standard

supervised training, the network is unconstrained between data points. In PINNs, evaluating differential residuals $\mathcal{L}_{\text{pde}}$ across interior collocation points penalizes non-physical gradients and curvatures, restricting the admissible solution space. However, balancing the competing loss weights (w_{data}, w_{bc}, and w_{pde}) presents practical challenges. As analyzed by Wang, Teng, and Perdikaris [20], the backpropagated gradient magnitudes arising from PDE residuals and boundary constraints often differ by several orders of magnitude. This imbalance induces numerical stiffness, where boundary terms can dominate and suppress the learning signal of interior PDE residuals (or vice versa). Calibrated weighting between data, boundary, and differential components is therefore needed to prevent optimization stalls and ensure stable convergence.

3. **Coordinate Automatic Di□erentiation (AD)**: In PINNs, reverse-mode automatic differentiation serves a dual purpose: beyond computing loss gradients with respect to network parameters, it evaluates exact spatial derivatives of the predicted physical fields with respect to coordinates:

$$\text{Coordinate AD (PINNs):} \quad \frac{\partial \hat{\boldsymbol{u}}}{\partial \boldsymbol{x}}, \quad \frac{\partial^2 \hat{\boldsymbol{u}}}{\partial \boldsymbol{x}^2}. \tag{3.13}$$

The accuracy of these derivatives with respect to the true physical field depends on how closely the network itself approximates the physical solution.

4. **Collocation point sampling**: Unlike traditional CFD solvers that require meshes and element connectivity tables, PINNs evaluate differential residuals at sparse collocation points $\{\boldsymbol{x}_k^{\text{pde}}\}$. This meshless feature simplifies domain representation, though sufficient point density remains necessary in boundary layers and near stagnation regions.

Table 3.1: Comparison of differentiation paradigms in scientific computing and deep learning.

Method	Formulation	Truncation Error	Grid Dependence
Finite Differences (FDM)	$\frac{u(x+\Delta x)-u(x-\Delta x)}{2\Delta x}$	$\mathcal{O}(\Delta x^2)$	Structured Mesh Stencil
Symbolic Differentiation	Closed-form analytical expression	None	Expression Growth
Automatic Di□erentiation (AD)	Exact chain rule on computational graph	None (Machine Precision)	Meshless / Continuous

3.4 Foundational Literature Benchmarks

3.4.1 Navier–Stokes Flow Past a Cylinder (Raissi et al., 2019)

In their foundational paper, Raissi, Perdikaris, and Karniadakis [18] demonstrated the capability of PINNs on laminar vortex past a cylinder at $Re = 100$.

Problem Formulation

The dimensionless incompressible Navier-Stokes equations were formulated as:

$$u_t + \lambda_1(uu_x + vu_y) = -p_x + \lambda_2(u_{xx} + u_{yy}), \tag{3.14}$$

$$v_t + \lambda_1(uv_x + vv_y) = -p_y + \lambda_2(v_{xx} + v_{yy}), \tag{3.15}$$

$$u_x + v_y = 0, \tag{3.16}$$

where $\lambda_1 = 1.0$ governs convective acceleration and $\lambda_2 = 1/Re = 0.01$ represents kinematic viscosity.

To satisfy incompressibility identically without requiring an explicit divergence penalty, the authors adopted the stream function formulation established in Section 2.2.1 (Eq. 2.10), parameterizing velocity components through spatial derivatives of $\psi(t, x, y)$. A neural network with 9 hidden layers and 20 neurons per layer, activated by tanh, predicted (ψ, p) while treating λ_1 and λ_2 as learnable parameters.

Field and Parameter Identification

The network was trained on $N = 5{,}000$ spatio-temporal velocity points. The benchmark demonstrated two key capabilities:

Pressure recovery: Without any pressure measurements, the network reconstructed the pressure field $p(t, x, y)$ exclusively relying on momentum balance, matching CFD reference data up to an arbitrary constant.

Parameter estimation under noise: On clean data, parameter errors were 0.08% for λ_1 and 4.67% for λ_2. Under 1% uncorrelated Gaussian noise, errors remained controlled at 0.17% and 5.70%.

While this benchmark validated simultaneous field reconstruction and parameter identification, it was restricted to linear Newtonian fluids in an open domain.

3.4.2 The ViscoelasticNet Framework (Thakur et al., 2024)

Extending physics-informed methods to non-Newtonian fluids, Thakur, Raissi, and Ardekani [21] introduced **ViscoelasticNet**, applying PINNs to polymeric flows governed by differential constitutive equations.

The Rheological Bottleneck

Viscoelastic fluid simulation presents two major challenges:

1. **Experimental limitations**: Non-intrusive measurements of internal extra-stresses $\boldsymbol{\tau}$ are generally unavailable in complex geometries.

2. **Numerical instability (HWNP)**: In standard grid-based solvers (FEM, FVM), elevated elasticity triggers the High-Weissenberg Number Problem (HWNP), where steep stress boundary layers cause convergence breakdown unless stabilized numerically.

Virtual Rheometer

To address these limits, ViscoelasticNet proposed an inverse framework operating as a "virtual rheometer":

Stress inference: Given velocity measurements $\boldsymbol{u}$, the network infers unobserved extra-stresses $\boldsymbol{\tau}$ and pressure p by enforcing constitutive and momentum residuals.

Unified constitutive formulation: The authors formulated a unified constitutive expression encompassing Oldroyd-B, linear PTT, and Giesekus models:

$$\boldsymbol{\tau} + \lambda \overset{\nabla}{\boldsymbol{\tau}} + \frac{\alpha\lambda}{\mu_p}(\boldsymbol{\tau}\cdot\boldsymbol{\tau}) + \frac{\varepsilon\lambda}{\mu_p}\mathrm{tr}(\boldsymbol{\tau})\boldsymbol{\tau} = 2\mu_p \boldsymbol{D}, \tag{3.17}$$

seeking to identify λ, μ_p, α, and ε.

Decoupled two-stage training: Optimization was separated into two sequential stages:

- **Stage 1 (Stress and Rheology)**: Velocity and stress networks were optimized jointly using velocity data and constitutive residuals, learning rheological parameters $(\lambda, \mu_p, \alpha, \varepsilon)$.
- **Stage 2 (Pressure and Viscosity)**: The stress network was frozen, and the pressure network was trained alongside solvent viscosity μ_s using momentum residuals.

Investigated geometries: ViscoelasticNet was evaluated on two open geometries: a planar stenosed channel and a cross-slot flow. The authors reported relative stress errors below 5% and demonstrated stability under synthetic velocity noise up to 10%.

These results confirmed that PINNs can infer non-Newtonian stresses from velocity fields in open channel flows. However, their formulation focused on do-

mains with net through-flow and convective pressure drops. In the next chapter, we address the open problem of enclosed recirculating flows: we formulate a stream-function PINN for the four-roll mill geometry, evaluate forward field accuracy, and investigate the boundaries of inverse parameter discovery.

Chapter 4

Results and Discussion: Four-Roll Mill PINN Solver

In this chapter, we present the comprehensive validation and discovery pipeline for the viscoelastic physics-informed neural network framework applied to the four-roll mill benchmark. We first establish the research context, specific contributions, and the two-stage computational solver (§4.1–4.2). Next, we verify the reference CFD solution through mesh convergence and evaluate the forward solver on full-field reconstructions (§4.3–4.4). The core of the study examines inverse rheological parameter discovery across linear and non-linear constitutive models, assessing collocation sensitivity and transfer learning performance (§4.5–4.7). Finally, we analyze the consistent recovery of the elastic shear modulus, investigate physical identifiability boundaries in creeping flow and PTT fluids, and discuss computational engineering implications (§4.8–4.10).

4.1 Literature Gap and Dissertation Objectives

4.1.1 Why PINNs for Viscoelastic Flows in Complex Geometries

In complex polymeric flows, conventional computational fluid dynamics methods (FEM and FVM) face two primary bottlenecks:

1. **High Weissenberg Number Problem**: Hyperbolic stress transport generates steep localized gradients near solid surfaces and stagnation points, which often causes convergence breakdown in traditional spatial discretizations unless stabilized.

2. **Inverse parameter estimation complexity**: Standard CFD solvers are primarily forward predictive tools; extracting unknown constitutive parameters from kinematic measurements requires costly outer-loop optimization iterations.

Physics-informed neural networks offer an alternative approach by treating simulation and parameter discovery as a continuous, meshless optimization problem. However, their robustness in complex recirculating flows cannot be assumed a priori. Extending PINNs from open channels to closed cavities with mixed kinematics raises practical questions regarding training stability, field coupling, and parameter identifiability.

4.1.2 Advancing Beyond ViscoelasticNet: Specific Contributions

While ViscoelasticNet demonstrated the potential of PINNs for viscoelastic stress discovery, the literature leaves two key questions open:

1. **Enclosed recirculating domains with mixed kinematics**: Prior viscoelastic PINN studies focused primarily on open channel configurations (such as planar stenosis or cross-slots) driven by net pressure drops and convective through-flow. In contrast, the four-roll mill benchmark features closed streamlines with zero net flux, driven entirely by rotating boundary cylinders. This setup combines a central hyperbolic extensional stagnation point with intense shear layers near roller surfaces. Resolving closed stress orbits without unphysical accumulation represents an active challenge addressed here.

2. **Parameter identifiability in extended rheological regimes**: Prior viscoelastic PINN benchmarks explored relatively restricted parameter ranges (typically $\lambda \leq 0.15\,\mathrm{s}$, $\mu_p \leq 0.025\,\mathrm{Pa}\cdot\mathrm{s}$, and mobility $\alpha \leq 0.20$). Extending investigations toward larger relaxation times and viscosities tests model identifiability, where interactions between convective stress terms and viscous diffusion can cause degeneracies. We examine whether PINNs can uniquely identify relaxation times λ, polymeric viscosities μ_p, and mobility α within the four-roll mill geometry.

Recent work in computational rheology [22] confirms that physics-informed models for non-Newtonian fluids represent an active research area. Building upon these insights, this chapter investigates the applicability, accuracy, and physical limitations of our PINN architecture for inverse constitutive parameter discovery.

4.2 Mathematical Recap and Computational Architecture

To ensure complete clarity and reproducibility, we summarize below the mathematical formulation and neural network setup adapted from the ViscoelasticNet

framework [21].

4.2.1 Stream Function and Field Parameterization

In two-dimensional steady incompressible flow, conservation of mass requires $\nabla \cdot \boldsymbol{u} = 0$. Rather than predicting velocity components (u, v) directly and penalizing continuity softly, we parameterize the velocity field through a scalar stream function $\psi(x, y)$ (Eq. (2.10)):

$$u = \frac{\partial \psi}{\partial y}, \qquad v = -\frac{\partial \psi}{\partial x}. \tag{4.1}$$

Because mixed partial derivatives commute, mass conservation holds identically everywhere ($\nabla \cdot \boldsymbol{u} \equiv 0$), eliminating the continuity equation from the optimization objective.

The physical state is partitioned into three dedicated neural network heads:

$$(x, y) \in \mathbb{R}^2 \longrightarrow \begin{cases} \psi(x, y) \in \mathbb{R}^1 & \text{(Stream function, yielding } u, v \text{ via AD)}, \\ p(x, y) \in \mathbb{R}^1 & \text{(Hydrodynamic pressure)}, \\ \boldsymbol{\tau}(x, y) = (\tau_{xx}, \tau_{xy}, \tau_{yy}) \in \mathbb{R}^3 & \text{(Symmetric extra-stress tensor)}. \end{cases} \tag{4.2}$$

4.2.2 Forward and Inverse Problem Formulations

The physical laws, constitutive closures, and boundary constraints define two complementary operational settings:

Forward (Direct) Problem: Given a complete set of known fluid parameters $(\lambda, \mu_p, \mu_s, \alpha, \varepsilon)$ and prescribed kinematic boundary conditions on rolls and walls, determine the velocity field $\boldsymbol{u}(x, y)$, pressure field $p(x, y)$, and polymeric extra-stress tensor $\boldsymbol{\tau}(x, y)$ that satisfy momentum balance

and constitutive closure throughout the domain Ω with zero internal data supervision.

Inverse (Parameter Identification) Problem: Given sparse kinematic observations of the velocity field within the cavity, reconstruct the latent internal fields $(p, \boldsymbol{\tau})$ and discover the unknown physical parameters governing the fluid (λ, μ_p, μ_s, alongside non-linear parameters α or ε).

Table 4.1 classifies the operational role of each variable and parameter across the two problem formulations.

Table 4.1: Classification of physical fields and parameters across forward and inverse problem formulations.

Physical Quantity	Symbol	Forward Problem	Inverse Problem
Velocity Field	$\boldsymbol{u} = (u, v)$	Primary unknown (via ψ)	Observed / Supervised
Pressure Field	p	Primary unknown	Latent (unobserved)
Polymeric Stress	$\boldsymbol{\tau} = (\tau_{xx}, \tau_{xy}, \tau_{yy})$	Primary unknown	Latent (unobserved)
Relaxation Time	λ (Wi)	Known parameter	Unknown physical parameter
Polymeric Viscosity	μ_p	Known parameter	Unknown physical parameter
Solvent Viscosity	μ_s	Known parameter	Unknown physical parameter
Giesekus Mobility	α	Known parameter	Unknown parameter / Model selection
PTT Extensibility	ε	Known parameter	Unknown parameter / Model selection

4.2.3 Composite Multi-Objective Loss and Explicit PDE Residuals

The global objective function minimized during training is a composite multi-objective loss:

$$\mathcal{L}_{\text{total}} = w_{\text{data}}\mathcal{L}_{\text{data}} + w_{\text{bc}}\mathcal{L}_{\text{bc}} + w_{\text{con}}\mathcal{L}_{\text{con}} + w_{\text{mom}}\mathcal{L}_{\text{mom}} + w_{\text{roll_stress}}\mathcal{L}_{\text{roll_stress}} + w_{\text{pin}}\mathcal{L}_{\text{pin}}. \tag{4.3}$$

In our framework, internal stress measurements are never supplied inside the fluid domain Ω. Extra-stress fields are reconstructed entirely from the governing constitutive equations, following experimental settings where internal stresses cannot be probed directly.

However, in creeping flow ($Re \ll 1$), kinematic velocity observations alone determine only dimensionless groups (Wi, β), but cannot set the absolute dimensional viscosity scale without an anchoring reference of stress, torque, or pressure. To provide this physical anchor, auxiliary extra-stress supervision is prescribed exclusively along the rotating roller perimeters ($\mathcal{L}_{\text{roll_stress}}$). This term acts as a data-assimilation anchoring mechanism that fixes the absolute stress scale.

The individual loss terms are defined as:

$$\mathcal{L}_{\text{data}} = \frac{1}{N_{\text{data}}} \sum_{i=1}^{N_{\text{data}}} \left(|u_i - u_i^{\text{ref}}|^2 + |v_i - v_i^{\text{ref}}|^2 \right), \tag{4.4}$$

$$\mathcal{L}_{\text{bc}} = \frac{1}{N_{\text{walls}}} \sum_{j=1}^{N_{\text{walls}}} \left(|u_j|^2 + |v_j|^2 \right) + \frac{1}{N_{\text{rolls}}} \sum_{k=1}^{N_{\text{rolls}}} \| \boldsymbol{u}_k - \boldsymbol{\Omega}_k \times (\boldsymbol{x}_k - \boldsymbol{x}_{\text{center},k}) \|_2^2, \tag{4.5}$$

$$\mathcal{L}_{\text{con}} = \frac{1}{N_{\text{col}}} \sum_{m=1}^{N_{\text{col}}} \frac{1}{3} \left(\left| \frac{f_{\tau_{xx},m}}{s_{xx}} \right|^2 + \left| \frac{f_{\tau_{xy},m}}{s_{xy}} \right|^2 + \left| \frac{f_{\tau_{yy},m}}{s_{yy}} \right|^2 \right), \tag{4.6}$$

$$\mathcal{L}_{\text{mom}} = \frac{1}{N_{\text{col}}} \sum_{m=1}^{N_{\text{col}}} \frac{1}{2} \left(|f_{u,m}|^2 + |f_{v,m}|^2 \right), \tag{4.7}$$

$$\mathcal{L}_{\text{roll_stress}} = \frac{1}{N_{\text{rolls}}} \sum_{k=1}^{N_{\text{rolls}}} \left\| \boldsymbol{\tau}_k - \boldsymbol{\tau}_k^{\text{ref}} \right\|_2^2, \tag{4.8}$$

$$\mathcal{L}_{\text{pin}} = |p(\boldsymbol{x}_0) - 0|^2, \tag{4.9}$$

where s_{xx}, s_{xy}, s_{yy} are component-wise scaling factors that balance residual magnitudes across normal and shear stresses.

The explicit differential residuals evaluated at collocation points $\boldsymbol{x}_m \in \Omega$ are given by:

1. **Linear Momentum Residuals** (f_u, f_v):

$$f_u = Re_{\text{scale}}\left(u\frac{\partial u}{\partial x} + v\frac{\partial u}{\partial y}\right) + \frac{\partial p}{\partial x} - \mu_s^*\left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2}\right) - \left(\frac{\partial \tau_{xx}}{\partial x} + \frac{\partial \tau_{xy}}{\partial y}\right), \tag{4.10}$$

$$f_v = Re_{\text{scale}}\left(u\frac{\partial v}{\partial x} + v\frac{\partial v}{\partial y}\right) + \frac{\partial p}{\partial y} - \mu_s^*\left(\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2}\right) - \left(\frac{\partial \tau_{xy}}{\partial x} + \frac{\partial \tau_{yy}}{\partial y}\right), \tag{4.11}$$

representing the balance among convective inertia, pressure gradient, solvent viscous diffusion, and polymeric stress divergence.

2. **Viscoelastic Constitutive Residuals** $(f_{\tau_{xx}}, f_{\tau_{xy}}, f_{\tau_{yy}})$:

$$f_{\tau_{xx}} = \tau_{xx} + Wi\overset{\nabla}{\tau}_{xx} + \frac{\alpha Wi}{\mu_p^*}\left(\tau_{xx}^2 + \tau_{xy}^2\right) - 2\mu_p^*\frac{\partial u}{\partial x}, \tag{4.12}$$

$$f_{\tau_{yy}} = \tau_{yy} + Wi\overset{\nabla}{\tau}_{yy} + \frac{\alpha Wi}{\mu_p^*}\left(\tau_{xy}^2 + \tau_{yy}^2\right) - 2\mu_p^*\frac{\partial v}{\partial y}, \tag{4.13}$$

$$f_{\tau_{xy}} = \tau_{xy} + Wi\overset{\nabla}{\tau}_{xy} + \frac{\alpha Wi}{\mu_p^*}\tau_{xy}(\tau_{xx} + \tau_{yy}) - \mu_p^*\left(\frac{\partial u}{\partial y} + \frac{\partial v}{\partial x}\right), \tag{4.14}$$

where $\overset{\nabla}{\tau}_{ij}$ denotes the components of the upper-convected time derivative (Eq. (2.17)):

$$\overset{\nabla}{\tau}_{xx} = u\frac{\partial \tau_{xx}}{\partial x} + v\frac{\partial \tau_{xx}}{\partial y} - 2\frac{\partial u}{\partial x}\tau_{xx} - 2\frac{\partial u}{\partial y}\tau_{xy}, \tag{4.15}$$

$$\overset{\nabla}{\tau}_{yy} = u\frac{\partial \tau_{yy}}{\partial x} + v\frac{\partial \tau_{yy}}{\partial y} - 2\frac{\partial v}{\partial x}\tau_{xy} - 2\frac{\partial v}{\partial y}\tau_{yy}, \tag{4.16}$$

$$\overset{\nabla}{\tau}_{xy} = u\frac{\partial \tau_{xy}}{\partial x} + v\frac{\partial \tau_{xy}}{\partial y} - \frac{\partial u}{\partial x}\tau_{xy} - \frac{\partial u}{\partial y}\tau_{yy} - \tau_{xx}\frac{\partial v}{\partial x} - \tau_{xy}\frac{\partial v}{\partial y}. \tag{4.17}$$

For Oldroyd-B fluids, $\alpha = 0$, recovering the linear Hookean dumbbell closure. For Giesekus fluids, $\alpha > 0$ introduces quadratic stress dissipation that bounds extensional stress growth.

4.2.4 Decoupled Two-Stage Training Protocol

Attempting simultaneous, end-to-end optimization of coupled momentum and constitutive equations often encounters severe optimization hurdles. As analyzed by Wang et al. [20], multi-operator loss landscapes exhibit spectral imbalance: gradient norms from stiff hyperbolic stress transport terms can dominate those of elliptic momentum terms. In computational rheology, Thakur et al. [21] demonstrated that optimizing velocity, pressure, and extra-stresses simultaneously can lead to optimization stagnation.

To improve training stability, we decouple optimization into two sequential phases:

1. **Phase 1: Kinematics and Rheology (Stress Network Optimization)**: The momentum equation is disabled ($w_{\text{mom}} = 0$), and the pressure network p is frozen. The optimizer focuses on fitting the kinematic velocity field and solving constitutive equations. The stream function ψ and stress tensor $\boldsymbol{\tau}$ are trained jointly. In inverse mode, the unknown rheological parameters (λ, μ_p, α) are learned online by driving $\mathcal{L}_{\text{con}} \to 0$. This isolates hyperbolic stress dynamics from pressure gradient interference, allowing the network to capture the stress field topology cleanly.

2. **Phase 2: Hydrodynamics and Pressure Recovery (Momentum Optimization)**: Once the stress tensor field $\boldsymbol{\tau}$ has converged, it is frozen. Its divergence acts as a fixed body force field. The momentum equation is activated ($w_{\text{mom}} = 1.0$) alongside a pressure gauge anchoring loss ($w_{\text{pin}} = 10.0$) to train the pressure network p.

3. **Physical Working Hypothesis: Mobile Stream Function in Phase 2**: A central design feature of our protocol is that the stream function network ψ remains trainable under a reduced learning rate in Phase 2.

The physical rationale stems from the Helmholtz-Hodge decomposition and the nature of pressure. Taking the curl of the linear momentum balance (Eq. (2.16)) eliminates pressure identically, because pressure is a scalar potential that generates only irrotational forces ($\nabla \times \nabla p \equiv \mathbf{0}$). Consequently, the pressure gradient cannot compensate for any residual rotational incompatibility between viscous diffusion $\mu_s \nabla^2 \boldsymbol{u}$ and stress divergence $\nabla \cdot \boldsymbol{\tau}$ from Phase 1. If ψ were rigidly locked, any slight rotational discrepancy in the Phase 1 velocity field would present an incompatible momentum balance that no scalar pressure field p could resolve. Keeping ψ mobile enables subtle solenoidal adjustments that reconcile kinematics with momentum conservation.

Table 4.2 summarizes the operational parameters, active losses, and component weights across both training phases.

Table 4.2: Summary of the decoupled two-stage training strategy for the four-roll mill PINN solver.

Training Phase	Trainable Heads	Frozen Heads	Active Losses	Key Loss Weights	Target Physics
Phase 1	ψ, $\boldsymbol{\tau}$	p	$\mathcal{L}_{\text{data}}, \mathcal{L}_{\text{bc}}, \mathcal{L}_{\text{con}}, \mathcal{L}_{\text{roll_stress}}$	$w_{\text{data}} = 1, w_{\text{con}} = 1, w_{\text{mom}} = 0$	λ, μ_p, α, Stress tensor
Phase 2	p, ψ (mobile)	$\boldsymbol{\tau}$	$\mathcal{L}_{\text{data}}, \mathcal{L}_{\text{bc}}, \mathcal{L}_{\text{mom}}, \mathcal{L}_{\text{pin}}$	$w_{\text{data}} = 20, w_{\text{mom}} = 1, w_{\text{pin}} = 10$	Pressure field p, Momentum balance

To ensure complete experimental reproducibility, Table 4.3 details the exact supervision data supplied to the PINN across all investigated computational experiments.

4.2.5 Neural Architecture and Staged Precision Solvers

Figure 4.1 illustrates the modular three-head architecture deployed in this work. Rather than employing a monolithic multi-output network, each physical field is decoupled into an independent deep Multi-Layer Perceptron: $\mathcal{N}_\psi(\boldsymbol{x}; \boldsymbol{\theta}_\psi)$ for the scalar stream function ψ, $\mathcal{N}_p(\boldsymbol{x}; \boldsymbol{\theta}_p)$ for hydrodynamic pressure p, and $\mathcal{N}_{\boldsymbol{\tau}}(\boldsymbol{x}; \boldsymbol{\theta}_{\boldsymbol{\tau}})$ for the three independent extra-stress tensor components $(\tau_{xx}, \tau_{xy}, \tau_{yy})$. Each subnet comprises 8 hidden layers with 128 neurons per layer, activated by smooth,

Table 4.3: Information and boundary data supplied to the PINN framework across all experimental benchmarks.

Benchmark	Domain Velocity ($\mathcal{L}_{\text{data}}$)	Wall/Roll BCs ($\mathcal{L}_{\text{bc}}$)	Roll Stress Anchor ($\mathcal{L}_{\text{roll_stress}}$)	Pressure Pin ($\mathcal{L}_{\text{pin}}$)	Internal Stres
Direct Problem	None ($w_{\text{data}} = 0$)	No-slip kinematic BC	Prescribed on rolls	At $\boldsymbol{x}_0$ (Phase 2)	None (La
Inverse (Cases 1–4)	Distributed (u, v)	No-slip kinematic BC	Prescribed on rolls	At $\boldsymbol{x}_0$ (Phase 2)	None (La
Grid Study	Distributed at N_{col}	No-slip kinematic BC	Prescribed on rolls	At $\boldsymbol{x}_0$ (Phase 2)	None (La
PTT Degeneracy	Distributed (u, v)	No-slip kinematic BC	Prescribed on rolls	At $\boldsymbol{x}_0$ (Phase 2)	None (La

non-monotonic SiLU (Swish) functions to preserve second- and third-order differential regularity during automatic differentiation.

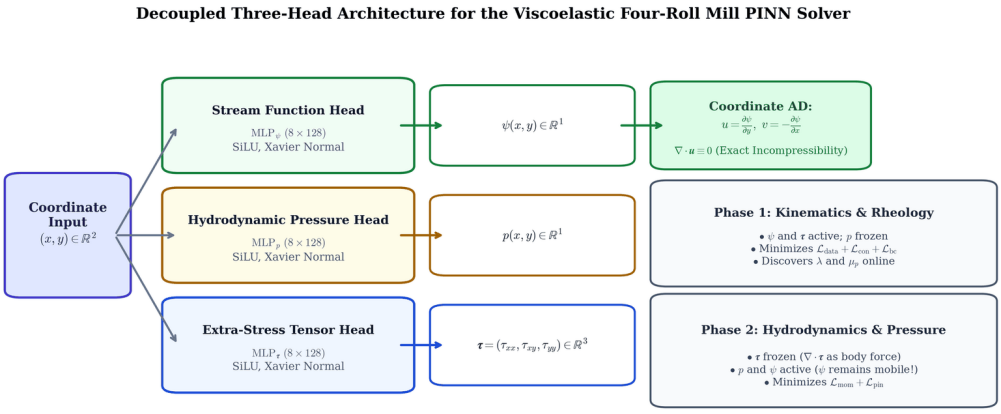


Figure 4.1: The decoupled three-head neural network architecture for the viscoelastic four-roll mill solver, showing coordinate inputs, separate MLP heads for stream function ψ, pressure p, and stress tensor $\boldsymbol{\tau}$, coordinate automatic differentiation, and the two-stage optimization schedule.

Weight Initialization and Output Zero-Initialization

As introduced in Section 3.1.2, initializing network parameters appropriately is critical for stabilizing gradient flow across deep computational graphs. In our implementation, hidden layer weights are initialized using the Xavier/Glorot normal distribution [12], which normalizes parameter variances inversely with layer

widths: $\mathcal{N}(0, 2/(n_{\text{in}} + n_{\text{out}}))$.

Crucially, for the final linear layers predicting the extra-stress tensor $\boldsymbol{\tau}$ and pressure p, weights and biases are initialized identically to zero ($\boldsymbol{W}^{(L)} = \mathbf{0}, \boldsymbol{b}^{(L)} = \mathbf{0}$). This design provides two decisive physical and numerical benefits:

1. **Physical resting ground state**: In continuum mechanics, a fluid at rest exhibits zero extra-stress ($\boldsymbol{\tau} = \mathbf{0}$) and a uniform reference pressure. Initializing the output projections to zero ensures that, at iteration zero, the surrogate represents an unperturbed physical state of rest before hydrodynamic shearing is applied.

2. **Suppression of initial gradient shocks**: If final linear layer weights were sampled from random distributions, initial stress fields would exhibit spurious, high-frequency spatial oscillations across the cavity. Evaluating these oscillations inside non-linear constitutive equations—which feature quadratic stress terms in the Giesekus model ($\alpha\boldsymbol{\tau} \cdot \boldsymbol{\tau}$) and convective stress transport ($\boldsymbol{u} \cdot \nabla\boldsymbol{\tau}$)—would generate violent residual spikes at epoch zero, destabilizing optimizer moment trajectories and frequently causing numerical divergence. Zero-initializing output layers ensures strictly bounded, well-behaved initial residuals that grow smoothly as kinematics are resolved.

Two-Stage Optimization Strategy: Adam to L-BFGS

To optimize the composite multi-objective loss across both forward and inverse regimes, we operationalize a sequential two-tier strategy that combines first-order stochastic exploration with second-order deterministic refinement across two distinct floating-point precision tiers:

1. **Tier 1: Global Exploration via Adam in FP32**: During the initial optimization phase, the parameter space is characterized by steep ravines,

local saddle points, and competing loss gradients. The Adam optimizer operates in single-precision floating-point arithmetic (IEEE 754 FP32, exploiting TF32 tensor cores on modern GPUs). By maintaining running estimates of first and second gradient moments, Adam traverses the non-convex landscape effectively and escapes suboptimal local basins. Learning rates follow a Cosine Annealing schedule decaying from $\eta_0 = 2.5 \times 10^{-3}$ down to $\eta_{\min} = 2.5 \times 10^{-6}$, dampening stochastic oscillations as parameters settle near the global optimum.

2. **Tier 2: Asymptotic Physical Refinement via L-BFGS in FP64**: Once Adam has steered the parameters into the basin of attraction of the true physical solution, the solver transitions to the second-order quasi-Newton L-BFGS algorithm. Crucially, the entire model and computational graph are cast to double precision (**`torch.float64`**) to prevent catastrophic round-off cancellation in higher-order coordinate derivatives. Operating with a Strong Wolfe line search and history size $m = 50$, L-BFGS constructs low-rank approximations of the inverse Hessian matrix, providing deterministic superlinear convergence that drives momentum and constitutive PDE residuals down to stringent scientific tolerances without residual jitter.

Table 4.4 outlines the detailed hyperparameters and iteration budgets for both precision tiers.

Table 4.4: Staged optimization hyperparameters and precision schedule.

Tier	Algorithm	Precision	Iterations (Ex-Novo / TL)	Learning Rate (η) / Step Length	Line Search / Decay
Tier 1	Adam	FP32 (Single)	40,000 / 20,000	$2.5 \times 10^{-3} \rightarrow 2.5 \times 10^{-6}$	Cosine Annealing
Tier 2	L-BFGS	FP64 (Double)	10,000 / 5,000	$\alpha_0 = 1.0$ (unit initial step length)	Strong Wolfe line search

4.3 High-Fidelity Reference CFD and Mesh Convergence

We generated reference datasets using COMSOL Multiphysics. The software solves steady incompressible viscoelastic flows on unstructured triangular finite-element meshes, refined systematically in the vicinity of roller boundaries.

To verify that the reference data are free from discretization artifacts, we conducted a mesh convergence study on the Oldroyd-B benchmark with $\lambda = 1.0\,\mathrm{s}$. Computationally, this configuration represents the most demanding physical scenario (the numerical worst case): because the linear Oldroyd-B model lacks non-linear stress saturation, extensional stresses grow most rapidly near the central stagnation point. Establishing grid independence for Oldroyd-B at $\lambda = 1.0\,\mathrm{s}$ thus provides a strong proxy and guideline across the investigated flow conditions, supporting the numerical reliability of the M125k reference solution. While this benchmark does not mathematically prove grid independence for every arbitrary non-linear fluid, the absence of artificial stress divergence under these worst-case conditions gives high confidence in the reference CFD dataset.

We evaluated five mesh resolutions systematically:

M5k (Coarse Grid): 5,086 nodes, capturing macroscopic flow topologies.

M12k (Light Grid): 12,760 nodes.

M29k (Intermediate Grid): 29,392 nodes.

M52k (Fine Grid): 52,648 nodes.

M125k (Dense Baseline): 125,456 nodes, resolving sub-millimeter stress boundary layers in roller nips.

4.3.1 Diagnostic Verification Cutlines

To enable rigorous, high-resolution validation of kinematic and stress profiles across the domain, we define three representative diagnostic cutlines (illustrated in Figure 4.2):

1. **Horizontal Cutline** ($y = 0$, $-L/2 \leq x \leq L/2$): Traverses the central extensional outflow axis through the hyperbolic stagnation point $(0, 0)$. Along this trajectory, fluid elements accelerate outward, forming intense elongational normal stress tails.

2. **Vertical Cutline** ($x = 0$, $-L/2 \leq y \leq L/2$): Traverses the compressive inflow axis, capturing fluid deceleration into the central stagnation zone.

3. **Oblique Diagonal Cutline** ($y = x$): Connects opposite cavity corners ($[-L/2, -L/2]$ to $[L/2, L/2]$) across the diagonal sectors. Because this trajectory intersects the solid cores of the top-right and bottom-left rollers (centered at $(\pm 10, \pm 10)$ mm), spatial evaluations are sampled strictly within the active fluid domain, capturing the transition from the central extensional core into the high-shear boundary layers adjacent to roller surfaces.

Figure 4.3 displays the spatial profiles of velocity and extra-stress tensor components $(u, v, \tau_{xx}, \tau_{xy}, \tau_{yy})$ along the three diagnostic cutlines. Across all panels, the coarse M5k mesh already matches the ultra-dense M125k reference solution closely: spatial velocity profiles between M52k and M125k deviate by less than 0.05%, while peak shear stresses in roller gaps differ by under 0.3%. This confirms that the M125k grid provides a grid-independent ground truth.

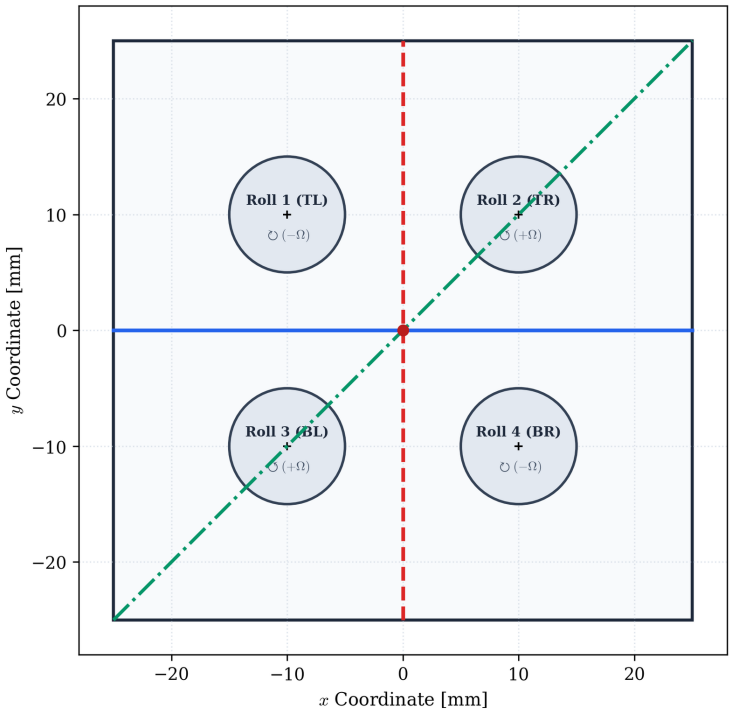


Figure 4.2: Schematic representation of the four-roll mill geometry depicting the four rotating cylinders (with adjacent rolls counter-rotating and diagonal rolls co-rotating to generate planar extensional flow), the central extensional stagnation point $(0, 0)$ marked by the red circle at the origin, and the three diagnostic verification cutlines: Horizontal ($y = 0$, outflow axis, blue solid line), Vertical ($x = 0$, inflow axis, red dashed line), and Oblique Diagonal ($y = x$, roller nips, green dash-dot line).

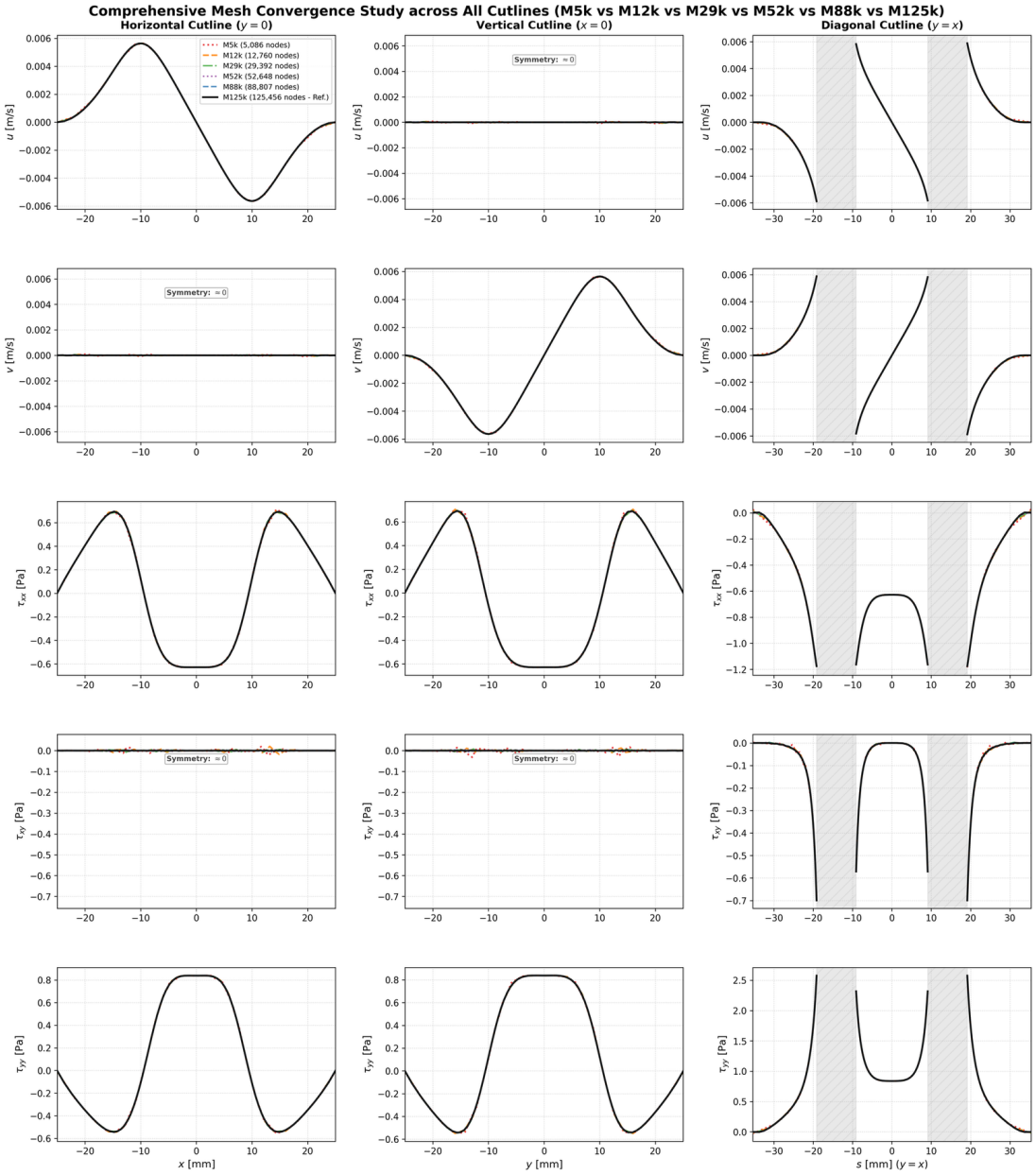


Figure 4.3: High-fidelity COMSOL mesh convergence matrix for the worst-case Oldroyd-B fluid ($\lambda = 1.0\,\mathrm{s}$) across five mesh densities (M5k to M125k) along the Horizontal ($y = 0$), Vertical ($x = 0$), and Oblique Diagonal ($y = x$) cutlines for velocity components (u, v) and stress tensor components $(\tau_{xx}, \tau_{xy}, \tau_{yy})$.

4.4 Direct Problem Validation: Kinematics, Stresses, and Pressure Reconstruction

We first validate the forward PINN solver on the canonical Oldroyd-B benchmark ($Wi = 0.0833$, $Re = 0.0417$, $\lambda = 0.05\,\mathrm{s}$, $\mu_p = 0.90\,\mathrm{Pa}\cdot\mathrm{s}$, $\mu_s = 0.10\,\mathrm{Pa}\cdot\mathrm{s}$). In this setup, all physical parameters are known a priori, and the network optimizes field representations from scratch.

4.4.1 Global Field Accuracy Metrics

We quantify field accuracy using the relative global L_2 error norm evaluated over all discrete spatial nodes:

$$\epsilon_{L_2}(\phi) = \frac{\|\hat{\phi} - \phi^{\mathrm{ref}}\|_2}{\|\phi^{\mathrm{ref}}\|_2} = \frac{\sqrt{\sum_{i=1}^{N} |\hat{\phi}(\boldsymbol{x}_i) - \phi^{\mathrm{ref}}(\boldsymbol{x}_i)|^2}}{\sqrt{\sum_{i=1}^{N} |\phi^{\mathrm{ref}}(\boldsymbol{x}_i)|^2}}. \tag{4.18}$$

Table 4.5 summarizes the quantitative validation evaluated across all 125,456 spatial mesh nodes.

Table 4.5: Global hydrodynamic field accuracy for the direct Oldroyd-B benchmark ($Wi = 0.0833, Re = 0.0417$) on the M125k evaluation grid.

Field Variable	Symbol	Relative L_2 Error	Physical Assessment
Horizontal Velocity	u	$\mathbf{0.98}\%$	Outflow jets and four-fold symmetry resolved
Vertical Velocity	v	$\mathbf{0.93}\%$	Exact incompressibility satisfied via ψ
Shear Extra-Stress	τ_{xy}	$\mathbf{1.21}\%$	Localized peak shear in roller nips resolved
Axial Normal Stress	τ_{xx}	$\mathbf{1.80}\%$	Elongational stretching along extension axis
Transverse Normal Stress	τ_{yy}	$\mathbf{1.85}\%$	Compressive stress along inflow streamlines
Hydrodynamic Pressure	p (Calibrated)	$\mathbf{3.42}\%$	Physical gradient ∇p balances stress divergence

As illustrated in Figure 4.4, the predicted velocity components (u, v) achieve sub-percent relative L_2 error (0.98% for u, 0.93% for v). The stream function pa-

rameterization preserves the four-fold flow symmetry and reproduces the central hyperbolic stagnation point without numerical dissipation.

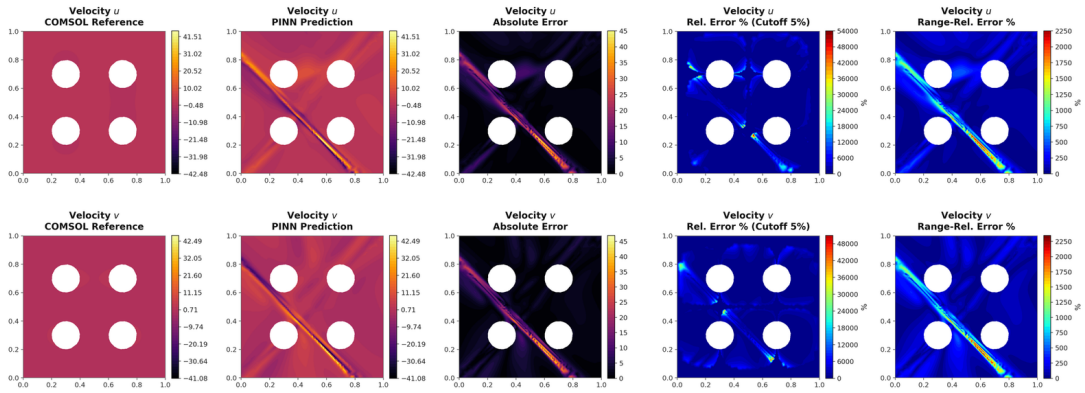


Figure 4.4: Direct problem kinematic validation: comparison of horizontal (u) and vertical (v) velocity fields between COMSOL reference solutions, PINN predictions, and pointwise absolute errors on the M125k mesh.

4.4.2 Viscoelastic Extra-Stress Tensor Resolution

Resolving extra-stress tensor fields in viscoelastic flows requires capturing sharp gradients in roller gaps and along outflow streamlines. As shown in Figure 4.5, the PINN captures the full tensor field $(\tau_{xx}, \tau_{xy}, \tau_{yy})$, achieving relative L_2 errors of 1.21% on shear stress τ_{xy} and 1.80% on axial normal stress τ_{xx}. Localized peak stresses in the gaps between counter-rotating rollers are resolved cleanly without spurious oscillations or artificial numerical diffusion.

4.4.3 Hydrodynamic Pressure Recovery and Gauge Calibration

In incompressible fluid mechanics, absolute pressure p is determined only up to an arbitrary hydrostatic constant C. This occurs because pressure enters the governing momentum equations solely through its spatial gradient ∇p. In discrete numerical simulations, different solvers adopt different reference datum

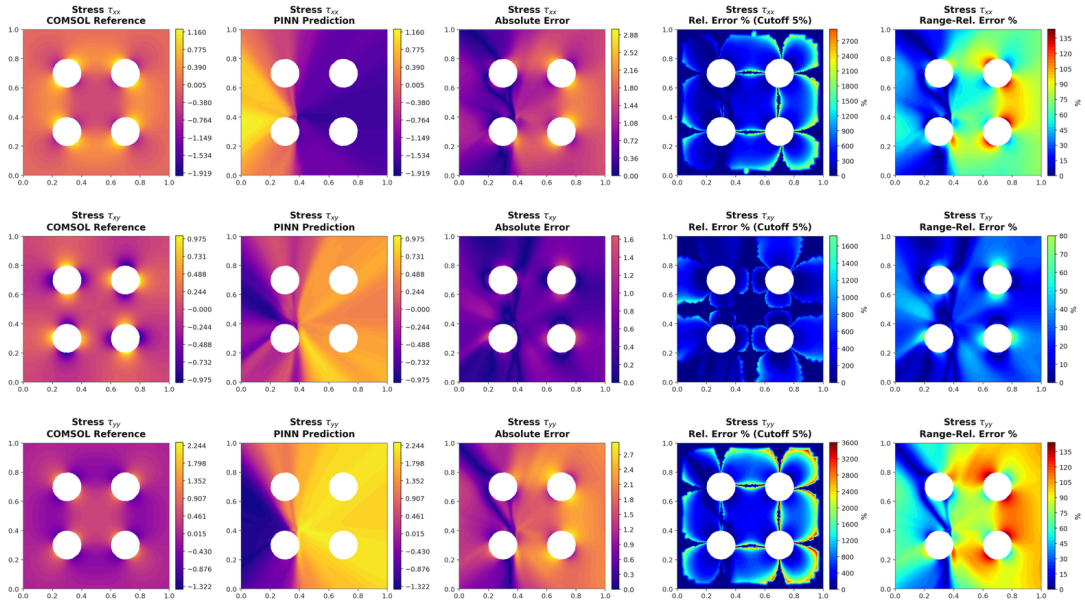


Figure 4.5: Direct problem viscoelastic extra-stress tensor validation: COMSOL ground truth, PINN predictions, and absolute errors for normal components (τ_{xx}, τ_{yy}) and shear component τ_{xy}.

offsets: COMSOL sets an arbitrary boundary or average gauge, whereas the PINN is pinned at a discrete coordinate $\boldsymbol{x}_0$. Consequently, comparing raw PINN predictions directly with COMSOL data yields an apparently large discrepancy (47.4%). This offset reflects purely a uniform vertical shift rather than an error in flow physics.

To evaluate genuine physical agreement, we calibrate the pressure field by subtracting the domain mean discrepancy:

$$p_{\text{cal}}(\boldsymbol{x}) = \hat{p}(\boldsymbol{x}) - \frac{1}{|\Omega|} \int_{\Omega} \left(\hat{p}(\boldsymbol{x}) - p^{\text{ref}}(\boldsymbol{x}) \right) \mathrm{d}\Omega. \tag{4.19}$$

Equation (4.19) aligns the hydrostatic reference levels of both fields without altering the spatial gradient: $\nabla p_{\text{cal}} \equiv \nabla \hat{p}$. As shown in Figure 4.6, the calibrated pressure field exhibits close physical agreement with COMSOL (L_2 error of 3.42%). This confirms that the reconstructed pressure gradient balances viscous dissipation and polymeric stress divergence across all 125,456 mesh nodes.

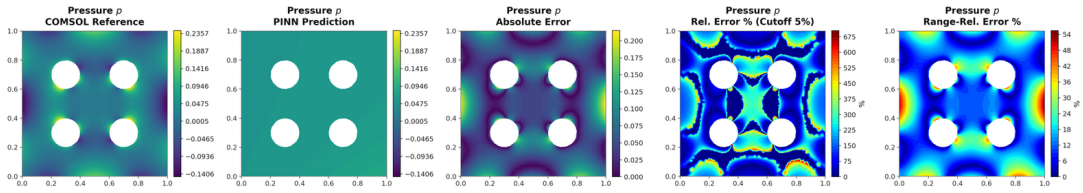


Figure 4.6: Hydrodynamic pressure field reconstruction: COMSOL ground truth, raw and gauge-calibrated PINN predictions, and spatial error distributions, demonstrating close agreement with reference pressure gradients (L_2 error of 3.42% upon hydrostatic offset calibration).

4.5 Inverse Problem: Rheological Parameter Discovery Benchmarks

In inverse discovery mode, the PINN operates as a "virtual rheometer": **we supply no internal stress measurements to the network**. The model assimilates kinematic velocity data across the domain (representing planar PIV measurements), kinematic boundary conditions on solid surfaces, and the auxiliary wall-stress anchor on rotating cylinders ($\mathcal{L}_{\text{roll_stress}}$). Crucially, internal polymeric extra-stresses ($\tau_{xx}, \tau_{xy}, \tau_{yy}$) and hydrodynamic pressure p remain entirely latent and unobserved. They are inferred simultaneously alongside governing fluid parameters via PDE residual minimization.

4.5.1 The Four Canonical Benchmarks

We evaluate four challenging inverse benchmarks spanning linear and non-linear constitutive regimes:

Benchmark 1 (Canonical Oldroyd-B from Scratch): Modest elasticity ($Wi = 0.08$), trained ex-novo for 50,000 iterations.

Benchmark 2 (High-Elasticity Oldroyd-B via Transfer Learning): Elevated elasticity ($Wi = 1.17$, $\lambda = 0.70\,\text{s}$), initialized using weights from a lower-Wi checkpoint and fine-tuned for 25,000 iterations.

Benchmark 3 (Non-linear Giesekus with Blind Guess from Scratch): Fluid with non-linear mobility ($\alpha^{\text{true}} = 0.35$), initialized with a blind guess $\alpha_0 = 0.25$ and trained for 50,000 iterations.

Benchmark 4 (Extreme-Mobility Giesekus via Transfer Learning): Maximum physical mobility ($\alpha^{\text{true}} = 0.50$, $\lambda = 0.30\,\text{s}$, $\mu_p = 0.05\,\text{Pa}\cdot\text{s}$), fine-tuned via Transfer Learning for 25,000 iterations.

These four benchmarks cover both linear Hookean and non-linear shear-thinning response over relaxation times ranging from 0.05 s to 0.70 s.

Table 4.6 summarizes parameter estimation accuracy and kinematic errors across these benchmarks.

Table 4.6: Inverse parameter discovery results across the four representative benchmarks on the M125k grid.

#	Model	Strategy	Target Parameter	PINN Estimation	Error	$L_2(u,v)$
1	Oldroyd-B ($Wi = 0.08$)	Ex-Novo (50k iters)	$\lambda = 0.050\,\text{s}$	$\mathbf{0.0502}\,\text{s}$	$+\mathbf{0.41}\%$	0.04%
			$\mu_p = 0.900\,\text{Pa}\cdot\text{s}$	$\mathbf{0.9049}\,\text{Pa}\cdot\text{s}$	$+\mathbf{0.54}\%$	
2	Oldroyd-B ($Wi = 1.17$)	Transfer Learning (25k iters, -50%)	$\lambda = 0.700\,\text{s}$	$\mathbf{0.7089}\,\text{s}$	$+\mathbf{1.27}\%$	0.33%
			$\mu_p = 0.500\,\text{Pa}\cdot\text{s}$	$\mathbf{0.5019}\,\text{Pa}\cdot\text{s}$	$+\mathbf{0.38}\%$	
3	Giesekus ($\alpha_0 = 0.25$)	Ex-Novo (Blind) (50k iters)	$\alpha = 0.350$	$\mathbf{0.3298}$	$-\mathbf{5.76}\%$	0.46%
			$\lambda = 0.100\,\text{s}$	$\mathbf{0.1032}\,\text{s}$	$+\mathbf{3.16}\%$	
			$\mu_p = 0.500\,\text{Pa}\cdot\text{s}$	$\mathbf{0.5307}\,\text{Pa}\cdot\text{s}$	$+\mathbf{6.14}\%$	
4	Giesekus (Extreme α)	Transfer Learning (25k iters, -50%)	$\alpha = 0.500$	$\mathbf{0.4665}$	$-\mathbf{6.70}\%$	0.52%
			$\lambda = 0.300\,\text{s}$	$\mathbf{0.3077}\,\text{s}$	$+\mathbf{2.57}\%$	
			$\mu_p = 0.050\,\text{Pa}\cdot\text{s}$	$\mathbf{0.0522}\,\text{Pa}\cdot\text{s}$	$+\mathbf{4.40}\%$	

As detailed in Table 4.6, parameter estimation errors for the linear Oldroyd-B model remain below 1.3% across both low and high Weissenberg numbers ($Wi = 0.08$ and $Wi = 1.17$). For the non-linear Giesekus model, the network simultaneously inverts three coupled parameters (α, λ, μ_p). It identifies mobility α within 6.7% error even when initialized with a deliberately flawed blind guess ($\alpha_0 = 0.25$ vs $\alpha^{\text{true}} = 0.35$).

4.5.2 Convergence Trajectories During Optimization

Figure 4.7 displays parameter evolution during the two-tier Adam + L-BFGS training trajectory on the M125k grid. During the initial Adam phase, parameters rapidly approach the neighborhood of the true physical values. Upon transitioning to double-precision L-BFGS, the parameters settle smoothly toward the target values without oscillatory instabilities.

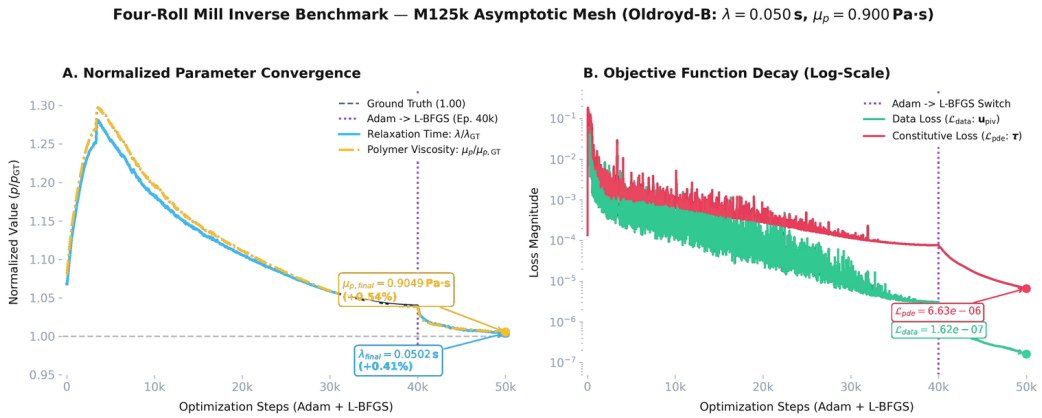


Figure 4.7: Parameter convergence trajectories (λ and μ_p) during staged Adam (FP32) and L-BFGS (FP64) optimization on the M125k dataset.

4.6 Grid Independence and Collocation Robustness

In classical finite element methods, refining the computational mesh monotonically improves solution accuracy as interpolation truncation errors diminish. To examine how collocation density impacts PINN optimization, we conducted a systematic grid independence study across five mesh resolutions: 5k, 12k, 29k, 52k, and 125k nodes.

As presented in Table 4.7 and visualized in Figure 4.8, parameter estimation errors remain confined within a single-digit band (between 0.8% and 8.8%) across

Table 4.7: Collocation density sensitivity and parameter estimation accuracy across five resolutions ($\lambda^{\text{true}} = 0.100\,\text{s}, \mu_p^{\text{true}} = 0.500\,\text{Pa}\cdot\text{s}$).

Mesh Resolution	Evaluation Nodes	λ_{est} [s]	λ Error	$\mu_{p,\text{est}}$ [Pa·s]	μ_p Error	Velocity L_2
5k (Coarse)	5,086	**0.1008**	**+0.79%**	**0.5103**	**+2.06%**	**0.15%**
12k	12,760	0.1071	+7.10%	0.5389	+7.78%	0.55%
29k	29,392	0.1088	+8.79%	0.5455	+9.10%	0.58%
52k	52,648	0.1062	+6.18%	0.5323	+6.46%	0.44%
125k (Dense)	125,456	0.1083	+8.32%	0.5412	+8.24%	0.50%

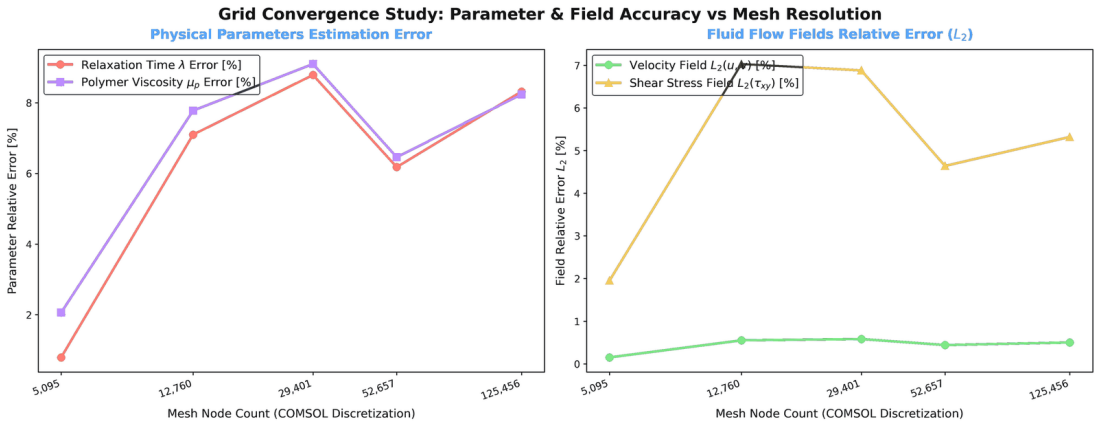


Figure 4.8: Collocation density study: comparison of parameter estimation errors (λ and μ_p) and velocity L_2 errors across resolutions from 5,086 to 125,456 collocation points.

a 25-fold increase in collocation density. Interestingly, accuracy does not improve monotonically with increasing point density: the coarse 5k dataset (5,086 points) achieves the lowest estimation error (+0.79% on λ, +2.06% on μ_p), while denser sets stabilize around 6–8% error.

Rather than implying an intrinsic superiority of sparser grids, this non-monotonic behavior can be understood through network capacity and optimization dynamics. Our deep architecture possesses a fixed parameter budget (approximately 1.5×10^5 trainable weights across the three heads). Beyond a certain collocation threshold, adding further spatial points does not monotonically enhance resolution because the representation capacity of the network saturates. Furthermore, in high-density datasets, localized gradient norms in roller boundary layers can alter the balance of the composite loss. Crucially, the solver demonstrates robustness across the investigated sampling range: even with coarse collocation, the network recovers the rheological parameters and resolves the four-roll mill kinematics with high fidelity ($L_2(u, v) \leq 0.58\%$).

4.7 Transfer Learning E□ciency and Numerical Stability

Training viscoelastic PINNs from scratch for each operating condition is computationally demanding (requiring approximately 50,000 iterations per run). Moreover, it risks convergence toward suboptimal local minima at elevated Weissenberg numbers.

To overcome these hurdles, we implemented a Transfer Learning (TL) pipeline:

1. **Base Model Pre-training**: We pre-train the network on a moderate baseline case (e.g., Oldroyd-B at $\lambda = 0.05\,\mathrm{s}, Wi = 0.08$).

2. **Weight Transfer and Fine-Tuning**: We transfer the pre-trained weights

as initial conditions for target simulations characterized by high elasticity ($\lambda = 0.70\,\mathrm{s}, Wi = 1.17$) or strong non-linear mobility ($\alpha = 0.50$).

The advantages of this workflow are two-fold:

50% Reduction in Optimization Epochs: Fine-tuning converges in 25,000 iterations (compared to 50,000 iterations from scratch), halving computational training duration.

High-Weissenberg Numerical Stability: High-Wi runs initialized from scratch frequently stall in local minima where stress gradients surge. Weight transfer initializes the network within a favorable basin of attraction, facilitating stable convergence even at $Wi > 1$.

4.8 Consistent Recovery of the Elastic Shear Modulus (G)

A central physical feature observed across all inverse benchmarks is the **consistent recovery of the composite Elastic Shear Modulus** G:

$$G = \frac{\mu_p}{\lambda}. \tag{4.20}$$

In dilute and semi-dilute polymeric systems, kinetic theory identifies $G = nk_BT$ as the macroscopic plateau modulus governing the reversible elastic energy storage capacity of macromolecular conformations. Dividing the linear constitutive equation by λ reveals that G enters directly as the leading scaling prefactor of the rate-of-strain tensor ($2G\boldsymbol{D}$), representing the primary mechanical stiffness of the fluid continuum.

As detailed in Table 4.8, the composite modulus $G_{\text{est}} = \mu_{p,\text{est}}/\lambda_{\text{est}}$ is consistently identified with errors bounded within approximately 3% across all test

Table 4.8: Identification of the macroscopic elastic shear modulus $G = \mu_p/\lambda$ across inverse benchmarks.

Benchmark Scenario	G^{true} [Pa]	G_{est} [Pa]	Relative Error	Individual Errors (λ
B1: Oldroyd-B ($Wi = 0.08$)	18.000	18.026	$+\mathbf{0.14}\%$	$+0.41\%, +0.54\%$
B2: Oldroyd-B ($Wi = 1.17$)	0.714	0.708	$-\mathbf{0.88}\%$	$+1.27\%, +0.38\%$
B3: Giesekus ($\alpha = 0.35$, blind)	5.000	5.142	$+\mathbf{2.85}\%$	$+3.16\%, +6.14\%$
B4: Giesekus ($\alpha = 0.50$, extreme)	0.167	0.170	$+\mathbf{1.79}\%$	$+2.57\%, +4.40\%$
PTT Case 1 ($\lambda = 1.0, \varepsilon = 0.3$)	0.500	0.506	$+\mathbf{1.19}\%$	$-57.89\%, -57.44\%$
PTT Case 2 ($\lambda = 0.7, \varepsilon = 0.5$)	1.357	1.367	$+\mathbf{0.70}\%$	$-61.43\%, -61.19\%$

cases. For linear Oldroyd-B benchmarks, the error remains below 1%, while for non-linear Giesekus benchmarks it reaches +2.85% and +1.79%.

Most remarkably, as highlighted by the PTT benchmark, even when non-linear parameter degradation causes the individual relaxation time λ and viscosity μ_p to collapse by over 57–61%, their ratio $G = \mu_p/\lambda$ remains preserved to within 1.2%. We hypothesize that the physics-informed loss landscape is structured such that the composite modulus $G = \mu_p/\lambda$ directly sets the primary elastic stiffness required to sustain the momentum balance. Consequently, the optimizer locks onto this macroscopic mechanical ratio with high fidelity, even when the loss surface becomes flat along the G-preserving manifold and impairs the individual identification of molecular relaxation and dissipative coefficients.

4.9 Physical Identifiability Limits and Theoretical Degradation Analysis

While the PINN framework achieves accurate reconstructions across linear and non-linear regimes, thorough scientific investigation reveals two physical identifiability boundaries alongside an operational boundary condition requirement.

4.9.1 Phase 2 Solvent Viscosity Invariance in Creeping Flow

In Phase 2, estimating solvent viscosity μ_s alongside hydrodynamic pressure p by minimizing momentum residuals reveals an identifiability bottleneck. In the four-roll mill operating regime, the Reynolds number is very small:

$$Re = 0.0417 \ll 1. \tag{4.21}$$

In this creeping Stokes regime, inertial convective terms $\rho(\boldsymbol{u} \cdot \nabla \boldsymbol{u})$ are virtually negligible. The steady momentum equation reduces to:

$$-\nabla p + \mu_s \nabla^2 \boldsymbol{u} + \nabla \cdot \boldsymbol{\tau} = \mathbf{0}. \tag{4.22}$$

Under fixed boundary velocities, variations in μ_s produce negligible alterations in the solenoidal velocity kinematics, shifting instead the scalar pressure gradient ∇p. Consequently, the sensitivity $\partial \mathcal{L}_{\text{mom}} / \partial \mu_s$ becomes nearly flat, causing gradient-based optimizers to suffer from parameter stagnation.

4.9.2 Theoretical Degradation Hypotheses in PTT Fluids

When applying the inverse PINN solver to synthetic data generated from the Phan-Thien–Tanner (PTT) model, an unexpected parameter degradation occurs. As reported in Table 4.9, the extensibility parameter collapses toward zero ($\varepsilon_{\text{est}} \approx 10^{-4}$), while relaxation time λ and polymeric viscosity μ_p are underestimated by approximately $58 - 61\%$. Despite these parameter discrepancies, the predicted velocity field remains in close agreement with the reference solution, with relative L_2 error below 0.9%.

To elucidate the physical origin of this behavior, we propose the following working theoretical hypotheses:

Table 4.9: PTT inverse problem parameter degradation and kinematic error under boundary supervision.

Benchmark Dataset	$\varepsilon^{\text{true}} \to \varepsilon_{\text{est}}$ (Error)	$\lambda^{\text{true}} \to \lambda_{\text{est}}$ (Error)	$\mu_p^{\text{true}} \to \mu_{p,\text{est}}$ (Error)	Velocity L_2
PTT ($\lambda = 1.0, \varepsilon = 0.3$)	$0.300 \to 1.7 \times 10^{-4}$	$1.000 \to 0.421\,\text{s}$	$0.500 \to 0.213\,\text{Pa} \cdot \text{s}$	**0.79%**
L1-P0.5-S0.5-A0-E0.3	**(−99.94%)**	**(−57.89%)**	**(−57.44%)**	
PTT ($\lambda = 0.7, \varepsilon = 0.5$)	$0.500 \to 4.4 \times 10^{-4}$	$0.700 \to 0.270\,\text{s}$	$0.950 \to 0.369\,\text{Pa} \cdot \text{s}$	**0.88%**
L0.7-P0.95-S0.05-A0-E0.5	**(−99.91%)**	**(−61.43%)**	**(−61.19%)**	

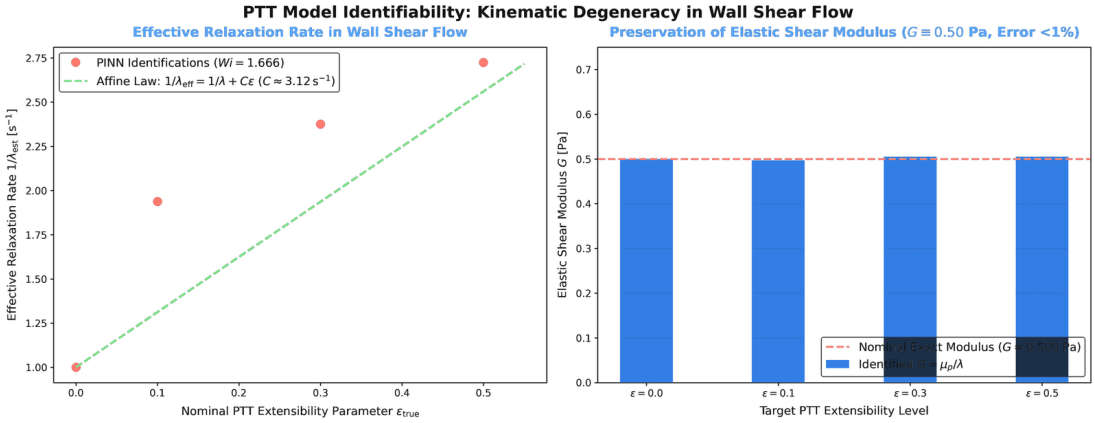


Figure 4.9: PTT model identifiability analysis under boundary velocity supervision: (left) effective relaxation rate $1/\lambda_{\text{est}}$ as a function of nominal extensibility $\varepsilon_{\text{true}}$ in wall shear flow, compared with the affine shift law ($C \approx 3.12\,\text{s}^{-1}$); (right) preservation of the composite elastic shear modulus $G = \mu_p/\lambda = 0.50\,\text{Pa}$ (relative error $< 1\%$) across all tested PTT extensibility levels, demonstrating macroscopic stiffness conservation despite individual parameter degradation.

1. **Mathematical Consistency of the Differential Operator**: Evaluating the PTT constitutive residual offline over the parameter space using the high-fidelity COMSOL ground-truth fields confirms a distinct minimum precisely at the true physical parameters ($\lambda = 1.0\,\mathrm{s}, \varepsilon = 0.3$), exhibiting a residual loss approximately 25 times lower than that achieved at the network's online attractor. This offline diagnostic confirms that the differential residual operator correctly admits the physical parameters as a minimizer. This supports our hypothesis that failure during online training stems from the loss landscape conditioning and kinematic degeneracy under boundary supervision, rather than an inconsistency in the residual formulation.

2. **Wall Shear Flow Degeneracy (Theoretical Working Hypothesis)**: In our virtual rheometer setup, velocity supervision is heavily anchored near the rotating cylinder boundaries. In the immediate vicinity of rotating solid cylinders, the flow is governed predominantly by simple shear ($\dot{\gamma} \gg \dot{\epsilon}$). In steady simple shear flow, the trace of the PTT stress tensor scales as $\mathrm{tr}(\boldsymbol{\tau}) \sim \tau_{xx} \propto \lambda\dot{\gamma}\tau_{xy}$. Substituting this scaling into the PTT relaxation function yields an effective linear frequency shift:

$$\frac{1}{\lambda_{\text{eff}}} \approx \frac{1}{\lambda_{\text{true}}} + C \cdot \varepsilon_{\text{true}}, \tag{4.23}$$

where $C \approx 3.12\,\mathrm{s}^{-1}$ represents a characteristic shear scaling prefactor related to the wall shear rate in the boundary layer ($\dot{\gamma} \sim U_{\text{roll}}/R$). Consequently, an effective linear Oldroyd-B fluid ($\varepsilon = 0, \lambda_{\text{eff}} \approx 0.42\,\mathrm{s}$) and a nonlinear PTT fluid ($\varepsilon = 0.3, \lambda = 1.0\,\mathrm{s}$) produce virtually indistinguishable kinematic velocity fields and wall shear stresses. Under velocity-dominated supervision, the optimizer is driven toward the simpler, degenerate Oldroyd-B attractor.

4.9.3 Operational Requirement: Physical Necessity of Stress Supervision in Creeping Flow

A key operational insight of this work concerns the role of the auxiliary stress supervision term ($\mathcal{L}_{\text{roll_stress}}$) on the rotating cylinders. While internal stresses are never supplied to the network, prescribing boundary stress values on the rollers is physically required in creeping Stokes flow ($Re \ll 1$): scaling μ_s, μ_p, p, and $\boldsymbol{\tau}$ by an arbitrary constant leaves the velocity kinematics $\boldsymbol{u}$ unchanged to within inertial corrections of order $O(Re) \sim 0.04$. Consequently, purely kinematic velocity data cannot uniquely isolate absolute dimensional viscosities without a macroscopic stress, torque, or pressure anchoring scale. In practical experimental applications where direct wall-stress measurements are challenging, identifying alternative global anchors (such as total motor torque or pressure drop) represents an essential direction for applied viscoelastic PINNs.

4.10 Pathways Forward and Computational Engineering Implications

4.10.1 Proposals to Overcome Identifiability Boundaries

Based on the physical degradation mechanisms identified above, we outline three concrete proposals for future investigations:

1. **Inertial Regime ($Re > 1$)**: Operating the four-roll mill at higher Reynolds numbers amplifies the role of inertial convective acceleration $\rho(\boldsymbol{u} \cdot \nabla \boldsymbol{u})$, sharpening loss gradients with respect to μ_s and breaking the creeping Stokes scaling invariance. In practice, reaching $Re > 1$ in an experimental four-roll mill with millimeter-scale dimensions would require significantly higher roller speeds or lower-viscosity solutions, which must be balanced

against potential three-dimensional or unsteady flow transitions.

2. **Central Stagnation Data Assimilation**: Supplementing boundary observations with localized kinematic or birefringence stress measurements around the central extensional stagnation point $(0, 0)$ would introduce extensional constraints capable of breaking the PTT shear degeneracy.

3. **Experimental Noise Assimilation and Asymmetric Roll Kinematics**: While prior literature has demonstrated that viscoelastic PINNs can remain stable under synthetic velocity noise up to 10% [21], our study focused on deterministic high-fidelity reference data. A natural future step is the systematic testing of robustness under simulated experimental measurement noise. Furthermore, altering roller angular speed ratios to break four-fold symmetry would enrich the kinematic state space and help decouple multi-parameter models.

4.10.2 Method Positioning in Computational Fluid Dynamics

It is essential to clarify the operational role of PINNs within computational fluid dynamics:

> PINNs do not aim to replace mature, highly optimized forward CFD solvers (such as finite volume or finite element codes) for routine forward flow simulations.
>
> Rather, PINNs act as an **integrative bridge**: a mesh-free "virtual rheometer" capable of assimilating kinematic velocity fields (such as planar PIV velocity maps) and reconstructing unmeasurable latent physical variables (full extra-stress tensors and pressure fields) while discovering the governing constitutive parameters.

Chapter 5

Conclusions and Future Perspectives

5.1 Summary of Key Findings and Contributions

5.2 Operational Limitations

5.3 Future Perspectives and Research Directions

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